L2 Geometry Workbook for Profile Sieve EL

Main-Body Trial: Geometry-First Proof With Detailed Appendix

Trial structure: main proof first, detailed machinery retained later

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Part I

Main Body Trial: Geometry-First Proof

1 Model Setup: Full Geometry in $L^2(P_T)$ and $L^2(P)$

The purpose of this workbook is to rewrite the whole proof logic as Hilbert space geometry. The sample space is not first probability; it is first the finite empirical Hilbert space

$$L^2(P_T) = (\mathbb{R}^T, \langle a, b\rangle_T = T^{-1}a'b).$$

The population limit is the Hilbert space

$$L^2(P), \qquad \langle f, g\rangle_P = \mathbb{E}\{fg\}.$$

The model setup is the following chain:

$$\begin{aligned}
\boldsymbol{y} &= \beta_0\boldsymbol{x} + \boldsymbol{\mu}_0 + \boldsymbol{u} && \text{model in } L^2(P_T),\\
\mu_{0,t} &= m_{j_0(t)}(W_{t-1}), \quad j_0(t) = j \iff k_{j,0} < t \le k_{j+1,0} && \text{true nuisance vector,}\\
\mathcal{N}_{K,\Lambda,T} &= \operatorname{col}(\boldsymbol{Q}_{K,\Lambda}) && \text{nuisance subspace,}\\
\boldsymbol{M}_{K,\Lambda} &= I - \Pi_{K,\Lambda,T} && \text{residual-maker,}\\
S_T(\beta,\Lambda) &= \sqrt{T}\langle \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y}-\beta\boldsymbol{x}), \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\rangle_T && \text{projected score.}
\end{aligned}$$

Here $\boldsymbol{\mu}_0$ is exactly the stacked nonparametric nuisance part of the model. If there are no nuisance breaks, $\mu_{0,t} = m(W_{t-1})$. With breaks, $\mu_{0,t} = m_j(W_{t-1})$ on regime j, so $\boldsymbol{\mu}_0$ is not one stable function evaluated over the whole sample; it is the true piecewise nuisance surface evaluated observation by observation.

EL then uses the scalar products

$$Z_t(\beta,\Lambda) = \{\boldsymbol{M}_{K,\Lambda}(\boldsymbol{y}-\beta\boldsymbol{x})\}_t\{\boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\}_t$$

to form a one-dimensional empirical likelihood. EL is not the Hilbert projection; EL is the likelihood wrapper around the projected score.

> **Main-body proof map.** The front proof states the finite-sample geometry needed for the main theorem and then proves consistency and Wilks on a good event $\mathcal{G}_T$. The detailed projection algebra, stochastic certification, and older merge derivations are retained in the appendix-style supporting material. Probability builds $\mathcal{G}_T$; geometry carries the proof once $\mathcal{G}_T$ holds.

2 Unified Geometric Dictionary

This workbook uses one geometric dictionary throughout. The partition does not only index a design matrix. It indexes a nuisance subspace:

$$\mathfrak{N}_T : \Lambda \longmapsto \mathcal{N}_{K,\Lambda,T} = \operatorname{col}(\boldsymbol{Q}_{K,\Lambda}) \subset L^2(P_T).$$

Thus break search is a finite-dimensional subspace-selection problem. For each Λ, the profiled nuisance estimate is the foot of the perpendicular from $\boldsymbol{y}^{\circ}_{\beta} = \boldsymbol{y} - \beta\boldsymbol{x}$ to $\mathcal{N}_{K,\Lambda,T}$, and the profile residual

$$\boldsymbol{M}_{K,\Lambda}\boldsymbol{y}^{\circ}_{\beta}$$

lives in the normal space

$$\mathcal{N}^{\perp_T}_{K,\Lambda,T}.$$

The raw score direction $\boldsymbol{w}$ is also pushed into the same normal space:

$$\boldsymbol{M}_{K,\Lambda}\boldsymbol{w} \in \mathcal{N}^{\perp_T}_{K,\Lambda,T}.$$

Therefore inference for β takes place in the quotient geometry

$$L^2(P_T)/\mathcal{N}_{K,\Lambda,T} \quad \text{or equivalently in} \quad \mathcal{N}^{\perp_T}_{K,\Lambda,T}.$$

The projected moment is the inner product of two normal components:

$$\Psi_T(\beta,\Lambda) = \langle \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta\boldsymbol{x}), \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\rangle_T.$$

The empirical likelihood part adds a second geometry. For the scalar moment $g_t(\beta,\Lambda)$, define

$$C_T(\beta,\Lambda) = \left\{\pi \in \Delta_T : \sum_{t=1}^{T} \pi_t g_t(\beta,\Lambda) = 0\right\},$$

where $\Delta_T = \{\pi_t \geq 0, \sum_t \pi_t = 1\}$. EL is the intersection of the probability simplex with a moment hyperplane. The multiplier λ_T is the normal coordinate to that hyperplane, and Wilks is the local quadratic curvature of the logarithmic barrier around the uniform weight vector.

Finally, estimated partitions are compared in the topology the proof actually uses, not in operator norm. For a set of relevant directions $\mathcal{D}_T$, define

$$d_{\mathcal{D},T}(M_1, M_0) = \sup_{a,b\in\mathcal{D}_T} \left|T^{-1/2}a'(M_1 - M_0)b\right|.$$

The relevant directions are the few vectors that enter the score and derivative, such as $\boldsymbol{\mu}_0, \boldsymbol{u}, \boldsymbol{x}, \boldsymbol{w}$, and in the efficient theorem $\boldsymbol{w}^*$. The estimated-break proof requires

$$d_{\mathcal{D},T}(\boldsymbol{M}_{K,\widehat{\Lambda}}, \boldsymbol{M}_0) = o_p(1),$$

not $\|\boldsymbol{M}_{K,\widehat{\Lambda}} - \boldsymbol{M}_0\|_{op} = o_p(1)$.

3 Paper-Facing Procedure

The main theorem uses the null-imposed one-break proof object. For a tested value β, define

$$\boldsymbol{y}_{\beta} = \boldsymbol{y} - \beta\boldsymbol{x}.$$

All partition comparisons below are made with this $\boldsymbol{y}_\beta$. Thus, for a p-value at $H_0 : \beta = \beta_0$, set $\beta = \beta_0$. For a confidence set, repeat the same null-imposed profiling at each candidate β.

Let $q \in \{0, 1\}$. The value $q = 0$ is the stable-nuisance model and $q = 1$ is the one-nuisance-break model. The score weight is

$$w_t = \omega(Z_t),$$

where ω is arbitrary within the admissible score-weight class: boundedness or the required max-score condition, nondegenerate projected variance, and the projected LLN/CLT assumptions are imposed on the resulting residualized score. Bounded choices are convenient in simulations, but the theorem does not require a particular functional form such as a hyperbolic tangent.

Algorithm 1: Null-imposed one-break profile-sieve EL.

1. **Input.** Data $\{Y_t, X_{t-1}, W_{t-1}, Z_t\}_{t=1}^T$, tested value β, sieve dimension K, minimum segment length, candidate orders $q \in \{0, 1\}$, penalty scale $\kappa_T R_T$, and admissible score weight $w_t = \omega(Z_t)$.
2. **Search candidate nuisance subspaces.** For each $q \in \{0, 1\}$ and each admissible partition $\Lambda \in \mathcal{L}_q$, form the block-sieve matrix $\boldsymbol{Q}_{K,\Lambda}$, projection $\Pi_{K,\Lambda,T}$, and residual-maker $\boldsymbol{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$.
3. **Profile under the tested value.** Compute

$$\boldsymbol{u}_\Lambda(\beta) = \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta\boldsymbol{x}), \qquad \boldsymbol{v}_\Lambda = \boldsymbol{M}_{K,\Lambda}\boldsymbol{w},$$

   and the null-imposed profile criterion

$$RSS_K(\beta, \Lambda) = \|\boldsymbol{u}_\Lambda(\beta)\|_2^2.$$

4. **Select the partition for each order.** Set

$$\widehat{\Lambda}_q(\beta) \in \arg\min_{\Lambda \in \mathcal{L}_q} RSS_K(\beta, \Lambda).$$

5. **Select the order.** Use the workbook penalty

$$IC(q; \beta) = RSS_K\{\beta, \widehat{\Lambda}_q(\beta)\} + q\,\kappa_T R_T,$$

   and set

$$\widehat{q}(\beta) \in \arg\min_{q \in \{0,1\}} IC(q; \beta), \qquad \widehat{\Lambda}(\beta) = \widehat{\Lambda}_{\widehat{q}(\beta)}(\beta).$$

6. **Construct scalar EL scores.** With $\widehat{M}_\beta = \boldsymbol{M}_{K,\widehat{\Lambda}(\beta)}$, define

$$g_t(\beta, \widehat{\Lambda}) = \{\widehat{M}_\beta(\boldsymbol{y} - \beta\boldsymbol{x})\}_t\{\widehat{M}_\beta\boldsymbol{w}\}_t.$$

7. **Solve the scalar EL KKT equation.** Find $\lambda(\beta)$ such that

$$\sum_{t=1}^{T} \frac{g_t(\beta, \widehat{\Lambda})}{1 + \lambda(\beta) g_t(\beta, \widehat{\Lambda})} = 0, \qquad 1 + \lambda(\beta) g_t(\beta, \widehat{\Lambda}) > 0.$$

8. **Compute the EL ratio.**

$$\ell_T(\beta) = 2 \sum_{t=1}^{T} \log\{1 + \lambda(\beta) g_t(\beta, \widehat{\Lambda})\}.$$

   Reject $H_0 : \beta = \beta_0$ when $\ell_T(\beta_0)$ exceeds the χ^2_1 critical value.

Remark 1 (What Algorithm 1 does not claim)**.** Algorithm 1 is the proof object used by the main theorem. It does not assert that a particular computational shortcut, such as seeded binary segmentation or narrowest-over-threshold search, is part of the statistical contribution. Multiple-break searches can be used as exploratory diagnostics, but the paper-facing theorem and implementation in this workbook are the $q \in \{0, 1\}$ null-imposed procedure above.

Assumption 1 (Nuisance covariate, sieve, and break signal)**.** *The following primitive conditions are sufficient for the nuisance-covariate part of the high-level good event. They do not replace the good-event assumptions below; they explain how the W-driven block sieve can satisfy them.*

1. ***Predetermined design.*** *The lagged design variables are observed before the innovation:*

$$X_{t-1}, W_{t-1} \in \mathcal{F}_{t-1}, \qquad E(U_t \mid \mathcal{F}_{t-1}) = 0.$$

   The nuisance covariate is W_{t-1}. The score weight w_t is a separate object.

2. ***Stable nuisance-covariate law.*** *The process W_t has a stable marginal law P_W and satisfies the mixing or empirical-process conditions needed for uniform laws over the searched sieve and break classes. A primitive route is stationary alpha mixing with sufficiently fast decay.*

3. ***Sieve basis and envelope.*** *Let*

$$P_K(W) = (p_1(W), \ldots, p_K(W))'.$$

   The basis includes an intercept and has envelope

$$\zeta_K = \sup_w \|P_K(w)\|.$$

   The growth of K and ζ_K is slow enough for the projection and local-break perturbation bounds.

4. ***Uniform segment Gram stability.*** *For every admissible partition $\Lambda \in \mathcal{L}_T$, the block-sieve design satisfies*

$$0 < c \le \lambda_{\min}\{T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda}\} \le \lambda_{\max}\{T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda}\} \le C < \infty$$

with probability tending to one.

5. ***Regime-wise sieve approximation.*** *For each true regime $j = 0, \ldots, q_0$, m_j belongs to a smoothness class approximable by P_K. There exists $a_K \downarrow 0$ such that*

$$\max_{0\le j\le q_0} \inf_{c_j\in\mathbb{R}^K} \|m_j(W) - P_K(W)'c_j\|_{L^2(P_W)} \le a_K.$$

The induced true-partition remainder $\boldsymbol{r}_{\mu,T} = \boldsymbol{M}_0\boldsymbol{\mu}_0$ is score-negligible:

$$\|\boldsymbol{r}_{\mu,T}\|_T = o_p(1), \qquad \sqrt{T}\,|\langle \boldsymbol{r}_{\mu,T}, \boldsymbol{M}_0\boldsymbol{w}\rangle_T| = o_p(1),$$
$$P_T\{r^2_{\mu,T,t}(\boldsymbol{M}_0\boldsymbol{w})^2_t\} = o_p(1).$$

6. ***Break spacing and break signal.*** *In the paper-facing one-break case, $q_0 = 1$ and the true break k_0 satisfies*

$$\min(k_0, T - k_0) \ge \epsilon T$$

for some $\epsilon > 0$. The nuisance break strength is

$$\Delta_T^2 = \|m_1(W) - m_0(W)\|^2_{L^2(P_W)},$$

and the RSS drift generated by this signal dominates searched stochastic fluctuations outside the local window.

7. ***Rate compatibility.*** *With*

$$R_T = K + \log T + Ta_K^2, \qquad r_T = R_T/\Delta_T^2$$

in the fixed-strength route, or the corresponding weak-break radius in the detailed proof, assume

$$r_T/T \to 0, \qquad \zeta_K\sqrt{Kr_T/T} \to 0.$$

For unknown order, the workbook penalty obeys

$$K\log T = o(\kappa_T R_T), \qquad \kappa_T R_T = o(T\Delta_T^2).$$

4 Proof Architecture: Good-Event Geometry Before Probability

The main-body proof is organized around the geometric dictionary just defined. Instead of presenting many steps as stochastic lemmas first, it separates the argument into two layers:

deterministic geometry on a good event

+

stochastic proof that the good event
has probability $1 - o(1)$

The geometric layer is a finite-sample statement in $L^2(P_T)$. It uses only projection identities, KKT identities, distance comparisons, and deterministic size inequalities. The stochastic layer proves that the deterministic hypotheses of the geometric layer are satisfied with probability tending to one.

This is the main proof slogan:

$$\mathcal{G}_T \Longrightarrow \begin{cases} \text{moment bridge and consistency,} \\ \text{EL KKT quadratic reduction,} \\ \text{estimated-partition score and variance replacement,} \\ \text{Wilks reduction to a standardized score,} \\ \text{optional efficient-score linearization when } w = w^*. \end{cases}$$

The supporting stochastic appendix proves $\Pr(\mathcal{G}_T) \to 1$, and the remaining stochastic input is only the CLT/LLN for the oracle score.

5 The Good Event

Fix a deterministic sequence $\delta_T \downarrow 0$ and constants $0 < c < C < \infty$. The good event is written as a set of geometric regularity conditions:

$$\mathcal{G}_T = G_T^{\text{iso}} \cap G_T^{\text{approx}} \cap G_T^{\text{angle}} \cap G_T^{\text{local}} \cap G_T^{\text{select}} \cap G_T^{\text{convex}}.$$

These names are not cosmetic. They identify which geometric fact each stochastic lemma must certify.

1. **Approximate isometry: G_T^{iso}.** For all searched partitions $\Lambda \in \mathcal{L}_T$, the empirical Gram matrix of $\boldsymbol{Q}_{K,\Lambda}$ is nonsingular with eigenvalues in $[c, C]$. The empirical inner products appearing in Ψ_T, the derivative, and the variance differ from their population analogues by at most δ_T, uniformly over the needed finite-dimensional directions. This says the random $L^2(P_T)$ geometry is approximately isometric to the population geometry on the relevant subspaces.

2. **Small nuisance distance: G_T^{approx}.** With $\boldsymbol{r}_{\mu,T} = \boldsymbol{M}_0 \boldsymbol{\mu}_0$,

$$\|\boldsymbol{r}_{\mu,T}\|_T \le \delta_T, \qquad \sqrt{T}\,|\langle \boldsymbol{r}_{\mu,T}, \boldsymbol{v}_0 \rangle_T| \le \delta_T, \qquad P_T(r_{\mu,T,t}^2 v_{0,t}^2) \le \delta_T.$$

   This says the true nuisance surface lies close to the true break-aware sieve subspace, and its normal residue is negligible in the score direction.

3. **Angle and nondegeneracy: G_T^{angle}.** Let

$$A_T^0 = T^{-1/2} \boldsymbol{u}' \boldsymbol{v}_0, \qquad B_T^0 = T^{-1} \sum_{t=1}^{T} u_t^2 v_{0,t}^2.$$

   The score is bounded at the local KKT scale, $|A_T^0| \le C$, and the variance is nondegenerate, $B_T^0 \ge c$. At the population level, the transversality constant satisfies $|\Gamma| \ge c$. This is the finite-sample version of the statement that the residualized parametric direction has a nonzero angle with the chosen residualized score direction.

4. **Local subspace stability: G_T^{local}.** The estimated partition lies in the local window, $\widehat{\Lambda} \in \mathcal{N}_T(r_T)$, and throughout this window,

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} \sqrt{T}\,|\Psi_T(\beta_0, \Lambda) - \Psi_T(\beta_0, \Lambda_0)| \leq \delta_T,$$

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} |V_T(\Lambda) - V_T(\Lambda_0)| \leq \delta_T,$$

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} \sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \Lambda) - \Psi_T(\beta, \Lambda_0)| \leq \delta_T.$$

   Equivalently, the residual-makers are close in the directional topology needed by the proof, not necessarily in operator norm.

5. **Subspace selection: G_T^{select}.** Outside the local window, the projection-distance drift dominates the noise fluctuation:

$$\inf_{\Lambda \in \mathcal{L}_T \setminus \mathcal{N}_T(r_T)} \{RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0)\} > 0.$$

   For unknown order, every underfitted order has larger penalized criterion than q_0, and every overfitted order also has larger penalized criterion than q_0. This is the nearest-subspace selection event.

6. **Convex feasibility: G_T^{convex}.** The max condition holds,

$$\max_t |g_t(\beta_0, \widehat{\Lambda})|/\sqrt{T} \leq \delta_T,$$

   and the scalar convex hull of $\{g_t(\beta_0, \widehat{\Lambda}) : 1 \leq t \leq T\}$ contains zero in its interior. This says the simplex-hyperplane intersection defining EL is locally feasible and the logarithmic barrier expansion is valid.

The supporting stochastic appendix certifies

$$\Pr(\mathcal{G}_T) \to 1$$

for suitable $\delta_T \downarrow 0$, constants, and rates.

6 Geometry Theorem I: Consistency Is Deterministic on $\mathcal{G}_T$

Theorem 1 (Good-event consistency)**.** *Assume the population geometry from the detailed finite-sample geometry gives*

$$\Psi(\beta, \Lambda_0) = -(\beta - \beta_0)\Gamma, \qquad \Gamma \neq 0.$$

On $\mathcal{G}_T$, if $\widehat{\beta}_T \in \mathcal{B}$ satisfies

$$|\Psi_T(\widehat{\beta}_T, \widehat{\Lambda})| \leq \delta_T,$$

then

$$|\widehat{\beta}_T - \beta_0| \leq \frac{2\delta_T}{|\Gamma|}$$

for all large T. Hence consistency follows once $\Pr(\mathcal{G}_T) \to 1$.

Proof. On $\mathcal{G}_T$, the uniform moment bridge gives

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \widehat{\Lambda}) - \Psi(\beta, \Lambda_0)| \leq \delta_T.$$

Therefore

$$|\Psi(\widehat{\beta}_T, \Lambda_0)| \leq |\Psi(\widehat{\beta}_T, \Lambda_0) - \Psi_T(\widehat{\beta}_T, \widehat{\Lambda})| + |\Psi_T(\widehat{\beta}_T, \widehat{\Lambda})| \leq 2\delta_T.$$

The population geometry gives $|\Psi(\widehat{\beta}_T, \Lambda_0)| = |\widehat{\beta}_T - \beta_0||\Gamma|$, which proves the claim. □

7 Geometry Theorem II: EL Is Convex KKT Plus a Size Event

Theorem 2 (Good-event EL quadratic reduction)**.** *On $\mathcal{G}_T$, the scalar EL dual root at $(\beta_0, \widehat{\Lambda})$ exists and satisfies*

$$\lambda_T = \frac{P_T g_t(\beta_0, \widehat{\Lambda})}{P_T g_t(\beta_0, \widehat{\Lambda})^2} + o_{\mathcal{G}}(T^{-1/2}),$$

where $o_{\mathcal{G}}(\cdot)$ denotes a deterministic remainder controlled on $\mathcal{G}_T$. Moreover,

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{V_T(\widehat{\Lambda})} + o_{\mathcal{G}}(1).$$

Proof. The EL problem is the finite-dimensional convex simplex problem from the detailed finite-sample geometry. On $\mathcal{G}_T$, the scalar convex hull contains zero in its interior, so the KKT multiplier λ_T exists and satisfies

$$P_T\left\{\frac{g_t}{1 + \lambda_T g_t}\right\} = 0, \qquad 1 + \lambda_T g_t > 0.$$

The same event gives $P_T g_t = O(T^{-1/2})$, $P_T g_t^2 \geq c/2$, and $\max_t |g_t|/\sqrt{T} \leq \delta_T$. The deterministic KKT expansion from Lemma 7 then gives the multiplier expansion and

$$2\sum_t \log(1 + \lambda_T g_t) = \frac{T(P_T g_t)^2}{P_T g_t^2} + o_{\mathcal{G}}(1).$$

Since $P_T g_t = \Psi_T(\beta_0, \widehat{\Lambda})$ and $P_T g_t^2 = V_T(\widehat{\Lambda})$, the displayed formula follows. □

8 Geometry Theorem III: Directional Break Perturbations

The important v2 point is that the estimated-partition argument does not need operator-norm convergence of $\boldsymbol{M}_{K,\widehat{\Lambda}}$ to $\boldsymbol{M}_0$. Projection matrices for two nearby break partitions may differ substantially as operators. What matters is their action on the few directions entering the score:

$$\boldsymbol{y}^{\circ}_{\beta_0} = \boldsymbol{\mu}_0 + \boldsymbol{u}, \qquad \boldsymbol{x}, \qquad \boldsymbol{w}.$$

Proposition 1 (Good-event estimated-partition bridge). *On $\mathcal{G}_T$,*

$$\sqrt{T}\{\Psi_T(\beta_0, \widehat{\Lambda}) - \Psi_T(\beta_0, \Lambda_0)\} = o_{\mathcal{G}}(1),$$
$$V_T(\widehat{\Lambda}) - V_T(\Lambda_0) = o_{\mathcal{G}}(1),$$

and

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \widehat{\Lambda}) - \Psi_T(\beta, \Lambda_0)| = o_{\mathcal{G}}(1).$$

Proof. Because $\widehat{\Lambda} \in \mathcal{N}_T(r_T)$ on $\mathcal{G}_T$, the three displayed inequalities are immediate from the local directional replacement component of $\mathcal{G}_T$. The geometric reason this is plausible is local support: when a candidate break is h observations away from a true break, the regime labels differ only on a boundary block of size h. The nuisance part is a projection residual supported on that block plus the sieve remainder, and the stochastic part is the directional quantity $\boldsymbol{u}'(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}$. The supporting stochastic appendix proves these quantities are small with high probability; this proposition only uses the resulting good-event inequalities. $\square$

9 Geometry Theorem IV: Wilks Is a Two-Step Consequence

Theorem 3 (Good-event Wilks reduction). *On $\mathcal{G}_T$,*

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{(A_T^0)^2}{B_T^0} + o_{\mathcal{G}}(1),$$

where

$$A_T^0 = T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0, \qquad B_T^0 = T^{-1}\sum_{t=1}^{T} u_t^2 v_{0,t}^2.$$

Consequently, if the stochastic shell proves

$$A_T^0 \Rightarrow N(0, V), \qquad B_T^0 \to_p V > 0,$$

then

$$\ell_T(\beta_0, \widehat{\Lambda}) \Rightarrow \chi_1^2.$$

Proof. Theorem 2 gives

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{V_T(\widehat{\Lambda})} + o_{\mathcal{G}}(1).$$

The estimated-partition bridge replaces $\widehat{\Lambda}$ by Λ_0. At the true partition,

$$\sqrt{T}\Psi_T(\beta_0, \Lambda_0) = T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0 + T^{-1/2}\boldsymbol{r}'_{\mu,T}\boldsymbol{v}_0 = A_T^0 + o_{\mathcal{G}}(1),$$

and

$$V_T(\Lambda_0) = B_T^0 + o_{\mathcal{G}}(1).$$

Substitution gives the displayed good-event reduction. The final χ_1^2 conclusion is now purely stochastic Slutsky: geometry has reduced EL to the square of the standardized oracle score. $\square$

10 Geometry Theorem V: Break Search Is Subspace Selection

Proposition 2 (Good-event break localization and order selection)**.** *On the RSS separation component of $\mathcal{G}_T$, the known-order RSS estimator satisfies*

$$\widehat{\Lambda} \in \mathcal{N}_T(r_T).$$

On the unknown-order penalty component of $\mathcal{G}_T$,

$$\widehat{q} = q_0.$$

Proof. The finite-sample RSS identity from the detailed finite-sample geometry gives

$$RSS_K(\beta_0, \Lambda) = T\,\mathrm{dist}_T^2(\boldsymbol{y} - \beta_0\boldsymbol{x}, \mathcal{N}_{K,\Lambda,T}).$$

Thus break search is selection among candidate nuisance subspaces. On the RSS separation component of $\mathcal{G}_T$, every partition outside the local window has larger RSS than Λ_0, so an RSS minimizer must lie inside $\mathcal{N}_T(r_T)$. For order selection, the same deterministic comparison applies to the penalized criterion: underfitted orders have deterministic omitted-break distance loss larger than any penalty saving, and overfitted orders have penalty increases larger than any spurious distance gain. Therefore $\widehat{q} = q_0$ on the event. □

11 Optional Efficiency

The optional efficiency result also has a geometric good-event form. Add the efficient model and the efficient weight

$$w^* = X/\sigma^2(A), \qquad A = (J_0, W),$$

as in the optional theorem stated in the supporting material. Add the efficiency event $\mathcal{G}_T^{\mathrm{eff}}$:

$$\left\| \boldsymbol{M}_0\boldsymbol{w}^* - \left(\frac{\widetilde{X}_1}{\sigma^2(A_1)}, \ldots, \frac{\widetilde{X}_T}{\sigma^2(A_T)} \right)' \right\|_T \le \delta_T,$$

$$T^{-1/2}|\boldsymbol{r}'_{\mu,T}\boldsymbol{M}_0\boldsymbol{w}^*| \le \delta_T,$$

$$\sqrt{T}|\Psi_T^*(\beta_0, \widehat{\Lambda}) - \Psi_T^*(\beta_0, \Lambda_0)| \le \delta_T,$$

and

$$\left| \langle \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{x}, \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{w}^* \rangle_T - \langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{M}_0\boldsymbol{w}^* \rangle_T \right| \le \delta_T, \qquad \langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{M}_0\boldsymbol{w}^* \rangle_T \to I_{\mathrm{eff}} > 0.$$

Theorem 4 (Good-event efficient linearization)**.** *On $\mathcal{G}_T \cap \mathcal{G}_T^{\mathrm{eff}}$, if*

$$\Psi_T^*(\widehat{\beta}_T, \widehat{\Lambda}) = o_{\mathcal{G}}(T^{-1/2}),$$

then

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) = I_{\mathrm{eff}}^{-1} T^{-1/2} \sum_{t=1}^{T} \frac{U_t\{X_t - E(X_t \mid A_t)\}}{\sigma^2(A_t)} + o_{\mathcal{G}}(1).$$

Proof. For fixed Λ, $\Psi_T^*(\beta,\Lambda)$ is affine in β, with slope

$$-\langle \boldsymbol{M}_{K,\Lambda}\boldsymbol{x}, \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}^*\rangle_T.$$

Expanding the root at β_0 and using the derivative bridge gives

$$\sqrt{T}(\widehat{\beta}_T-\beta_0)=\{\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{M}_0\boldsymbol{w}^*\rangle_T\}^{-1}\sqrt{T}\Psi_T^*(\beta_0,\Lambda_0)+o_{\mathcal{G}}(1).$$

The efficient-score approximation event gives

$$\sqrt{T}\Psi_T^*(\beta_0,\Lambda_0)=T^{-1/2}\sum_{t=1}^{T}\frac{U_t\{X_t-E(X_t\mid A_t)\}}{\sigma^2(A_t)}+o_{\mathcal{G}}(1),$$

and the denominator converges to I_{eff}. This proves the result. The limiting normal distribution and efficiency bound are then supplied by the stochastic shell: the efficient-score CLT and the standard Hilbert-space information inequality. □

12 What Remains Stochastic

The trial main-body proof is geometric, but it does not pretend probability has disappeared. The genuinely stochastic tasks are exactly these:

1. uniform empirical Gram and inner-product laws, which put the sample projection geometry close to population geometry;
2. sieve approximation rates for μ_0, $E(w\mid A)$, and, in the efficient theorem, $E(w^*\mid A)$;
3. RSS concentration, which proves the distance gap for wrong partitions is not overwhelmed by noise;
4. break-date localization and unknown-order penalty probability statements;
5. martingale or mixing CLT for $T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0$, and the matching variance LLN;
6. max-score and convex-hull probability bounds for the scalar EL KKT root;
7. in the optional efficiency theorem, the efficient-score CLT and any plug-in variance-weight replacement.

Thus the main-body proof is:

$$\begin{gathered}\text{finite-sample geometry}\\ \Longrightarrow\\ \text{good-event finite-sample reductions}\\ \Longrightarrow\\ \text{supporting stochastic appendix proves }\Pr(\mathcal{G}_T)\to 1\\ \Longrightarrow\\ \text{CLT gives Wilks or efficiency.}\end{gathered}$$

The stochastic lemmas certify the good event rather than carrying the main logical burden.

13 Minimal Final Statement

The complete proof is governed by one geometric object: the subspace-valued map

$$\Lambda \longmapsto \mathcal{N}_{K,\Lambda,T} \subset L^2(P_T).$$

For a trial β, profiling is the orthogonal projection of $\boldsymbol{y} - \beta\boldsymbol{x}$ onto this nuisance subspace. The profiled residual and the score direction both live in the normal space $\mathcal{N}_{K,\Lambda,T}^{\perp_T}$, so the projected moment is

$$\Psi_T(\beta, \Lambda) = \langle \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta\boldsymbol{x}), \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\rangle_T.$$

Population identification is the corresponding normal-space statement in $L^2(P)$: the nuisance disappears under M_0^P, and transversality

$$\Gamma = \langle M_0^P X, M_0^P w\rangle_P \neq 0$$

forces the population moment line to cross zero only at β_0.

Break search is nearest-subspace selection:

$$RSS_K(\beta, \Lambda) = T \operatorname{dist}_T^2(\boldsymbol{y} - \beta\boldsymbol{x}, \mathcal{N}_{K,\Lambda,T}).$$

A wrong break chooses the wrong subspace. A stable no-break nuisance space can leave a nonzero normal component $\boldsymbol{M}_{\mathrm{st}}\boldsymbol{\mu}_0$; false predictivity is the nonzero angle between this omitted nuisance residue and the residualized score direction.

EL adds convex geometry. The empirical likelihood feasible set is

$$C_T(\beta, \Lambda) = \left\{ \pi \in \Delta_T : \sum_t \pi_t g_t(\beta, \Lambda) = 0 \right\},$$

the intersection of the probability simplex with a moment hyperplane. The EL multiplier is the normal coordinate to that hyperplane, and the Wilks expansion is the local quadratic curvature of the logarithmic barrier around uniform weights.

The stochastic lemmas certify the good event

$$\mathcal{G}_T = G_T^{\mathrm{iso}} \cap G_T^{\mathrm{approx}} \cap G_T^{\mathrm{angle}} \cap G_T^{\mathrm{local}} \cap G_T^{\mathrm{select}} \cap G_T^{\mathrm{convex}}, \qquad \Pr(\mathcal{G}_T) \to 1.$$

On this event the proof is deterministic geometry: consistency follows from the uniform moment bridge plus transversality; EL reduces to the square of the standardized projected score; estimated breaks can replace true breaks because $\widehat{M_T}$ is close to M_0 in the directional topology actually used by the score, not in global operator norm. The remaining probability is the oracle score CLT and variance LLN, giving

$$\ell_T(\beta_0, \widehat{\Lambda}) \Rightarrow \chi_1^2.$$

The optional efficiency theorem is separate. Under the conditional Gaussian location efficiency model and efficient weight $w^* = X/\sigma^2(A)$, the same normal-space projection gives

$$M_0^P w^* = \frac{X - E(X \mid A)}{\sigma^2(A)}$$

and hence the projected score equals the efficient score. With the efficient sieve approximation and estimated-partition derivative bridge,

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) = I_{\text{eff}}^{-1} T^{-1/2} \sum_{t=1}^{T} \frac{U_t\{X_t - E(X_t \mid A_t)\}}{\sigma^2(A_t)} + o_p(1).$$

Thus efficiency is a conditional add-on to the Wilks proof, not part of the minimal validity argument.

Part II
Detailed Supporting Material

Appendix Roadmap

The appendix retains the detailed machinery behind the front proof. The first block gives the finite-sample Hilbert and convex KKT identities. The second block records the stochastic bounds that certify $\Pr(\mathcal{G}_T) \to 1$. The third block keeps the older merge derivation and the optional efficiency argument in full detail.

A Finite-Sample Hilbert Space: Viewing the Entire Model in $L^2(P_T)$

A.1 The empirical space

Let

$$\Omega_T = \{1, \dots, T\}, \qquad P_T = T^{-1} \sum_{t=1}^{T} \delta_t.$$

Every sample vector $\boldsymbol{a} = (a_1, \dots, a_T)'$ is a function $a_T : \Omega_T \to \mathbb{R}$, with $a_T(t) = a_t$. The empirical inner product is

$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle_T = \int a_T b_T \, dP_T = T^{-1} \sum_{t=1}^{T} a_t b_t = T^{-1} \boldsymbol{a}'\boldsymbol{b}.$$

Thus all variables in the sample are elements of $L^2(P_T)$:

$$\boldsymbol{y}, \boldsymbol{x}, \boldsymbol{w}, \boldsymbol{u}, \boldsymbol{\mu}_0 \in L^2(P_T).$$

A.2 The full finite-sample model

The break-aware model is the vector identity in $L^2(P_T)$

$$\boldsymbol{y} = \beta_0 \boldsymbol{x} + \boldsymbol{\mu}_0 + \boldsymbol{u}, \tag{A.1}$$

where

$$\mu_{0,t} = \sum_{j=0}^{q_0} m_j(W_{t-1})\mathbf{1}\{k_{j,0} < t \le k_{j+1,0}\}.$$

So the full model says: $\boldsymbol{y}$ lies in the affine set

$$\beta_0\boldsymbol{x} + \boldsymbol{\mu}_0 + \boldsymbol{u}.$$

The inferential problem is to remove $\boldsymbol{\mu}_0$, not by estimating it as a main parameter, but by projecting away its nuisance subspace.

A.3 Candidate nuisance subspaces

For a candidate partition $\Lambda = (k_1, \ldots, k_q)$, define the block sieve matrix

$$\boldsymbol{Q}_{K,\Lambda} = [\boldsymbol{D}_0(\Lambda)\boldsymbol{P}, \ldots, \boldsymbol{D}_q(\Lambda)\boldsymbol{P}].$$

The sample nuisance subspace is

$$\mathcal{N}_{K,\Lambda,T} = \mathrm{col}(\boldsymbol{Q}_{K,\Lambda}) \subset L^2(P_T). \tag{A.2}$$

An element of this space has the form

$$g_T(t) = \sum_{j=0}^{q} \mathbf{1}\{t \in I_j(\Lambda)\}\boldsymbol{P}_K(W_{t-1})'c_j.$$

This is the finite-sample version of a regime-specific nonparametric nuisance function.

Let $j_\Lambda(t)$ denote the regime index induced by Λ. For a regime-wise nuisance tuple $h = (h_0, \ldots, h_q)$, define its stacked sample vector by

$$\boldsymbol{h}_{\Lambda,T} = \big(h_{j_\Lambda(1)}(W_0), \ldots, h_{j_\Lambda(T)}(W_{T-1})\big)'.$$

This vector is the empirical version of the candidate nuisance direction.

Lemma 1 (Uniform finite-sieve approximation in $L^2(P_T)$)**.** *Fix (K, Λ, T) and a nuisance class $\mathcal{H}$. For each $h \in \mathcal{H}$, let $\boldsymbol{h}_{\Lambda,T} \in L^2(P_T)$ be the stacked nuisance vector defined above. Define the finite-sieve approximation radius*

$$\rho_{K,T}(\mathcal{H}, \Lambda) = \sup_{h\in\mathcal{H}} \inf_{\gamma\in\mathbb{R}^{(q+1)K}} \|\boldsymbol{h}_{\Lambda,T} - \boldsymbol{Q}_{K,\Lambda}\gamma\|_T. \tag{A.3}$$

If $\rho_{K,T}(\mathcal{H}, \Lambda) = o_p(1)$, then the desired uniform $L^2(P_T)$ approximation holds:

$$\sup_{h\in\mathcal{H}} \|\boldsymbol{h}_{\Lambda,T} - \boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h)\|_T = \sup_{h\in\mathcal{H}} \|\boldsymbol{r}_{K,\Lambda,T}(h)\|_T = \rho_{K,T}(\mathcal{H}, \Lambda) = o_p(1). \tag{A.4}$$

Here, for each h,

$$\gamma_{K,\Lambda}(h) \in \operatorname*{argmin}_{\gamma\in\mathbb{R}^{(q+1)K}} \|\boldsymbol{h}_{\Lambda,T} - \boldsymbol{Q}_{K,\Lambda}\gamma\|_T^2,$$

$$\boldsymbol{h}_{K,\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h), \qquad \boldsymbol{r}_{K,\Lambda,T}(h) = \boldsymbol{h}_{\Lambda,T} - \boldsymbol{h}_{K,\Lambda,T}.$$

The ambient space is $L^2(P_T) \cong \mathbb{R}^T$*, so* $\boldsymbol{h}_{\Lambda,T}$*,* $\boldsymbol{h}_{K,\Lambda,T}$*, and* $\boldsymbol{r}_{K,\Lambda,T}(h)$ *are all* $T \times 1$ *vectors. The design matrix and coefficient vector have dimensions*

$$\boldsymbol{Q}_{K,\Lambda} \in \mathbb{R}^{T\times(q+1)K}, \qquad \gamma_{K,\Lambda}(h) \in \mathbb{R}^{(q+1)K}.$$

The finite nuisance space has dimension

$$d_{K,\Lambda,T} = \operatorname{rank}(\boldsymbol{Q}_{K,\Lambda}) \leq (q+1)K,$$

and

$$\boldsymbol{h}_{\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h) + \boldsymbol{r}_{K,\Lambda,T}(h), \tag{A.5}$$

where

$$\boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h) \in \mathcal{N}_{K,\Lambda,T} \subset L^2(P_T), \qquad \boldsymbol{r}_{K,\Lambda,T}(h) \in \mathcal{N}_{K,\Lambda,T}^{\perp_T} \subset L^2(P_T).$$

Thus $\boldsymbol{r}_{K,\Lambda,T}(h)$ *is a* $T \times 1$ *approximation-residual vector living in a* $(T - d_{K,\Lambda,T})$*-dimensional orthogonal complement, and*

$$\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{r}_{K,\Lambda,T}(h) = 0.$$

Proof. The column space $\mathcal{N}_{K,\Lambda,T} = \operatorname{col}(\boldsymbol{Q}_{K,\Lambda})$ is a finite-dimensional closed subspace of $L^2(P_T)$. Therefore the Hilbert projection theorem gives the closest finite-sieve approximation to every $\boldsymbol{h}_{\Lambda,T}$:

$$\Pi_{K,\Lambda,T}\boldsymbol{h}_{\Lambda,T} \in \mathcal{N}_{K,\Lambda,T}.$$

Because every element of $\mathcal{N}_{K,\Lambda,T}$ is $\boldsymbol{Q}_{K,\Lambda}\gamma$, the projection can be written as

$$\Pi_{K,\Lambda,T}\boldsymbol{h}_{\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}\gamma_{K,\Lambda}(h)$$

for any least-squares minimizer $\gamma_{K,\Lambda}(h)$. If $\boldsymbol{Q}_{K,\Lambda}$ is rank deficient, the coefficient vector need not be unique, but the fitted vector is unique.

Define

$$\boldsymbol{r}_{K,\Lambda,T}(h) = \boldsymbol{h}_{\Lambda,T} - \Pi_{K,\Lambda,T}\boldsymbol{h}_{\Lambda,T}.$$

The projection theorem gives

$$\langle \boldsymbol{r}_{K,\Lambda,T}(h), \boldsymbol{v}\rangle_T = 0 \qquad \text{for every } \boldsymbol{v} \in \mathcal{N}_{K,\Lambda,T},$$

which implies $\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{r}_{K,\Lambda,T}(h) = 0$. This proves the orthogonal decomposition (A.5) and the location $\boldsymbol{r}_{K,\Lambda,T}(h) \in \mathcal{N}_{K,\Lambda,T}^{\perp_T}$.

By the definition of orthogonal projection,

$$\|\boldsymbol{r}_{K,\Lambda,T}(h)\|_T = \inf_{\gamma \in \mathbb{R}^{(q+1)K}} \|\boldsymbol{h}_{\Lambda,T} - \boldsymbol{Q}_{K,\Lambda}\gamma\|_T.$$

Taking the supremum over $h \in \mathcal{H}$ gives exactly (A.4). Hence the mathematical task behind this lemma is to verify $\rho_{K,T}(\mathcal{H},\Lambda) = o_p(1)$ from the chosen basis, smoothness class, and growth rate of K. An arbitrary nuisance direction is not assumed to be exactly spanned by the sieve basis; it is approximated by its finite-sieve projection, and the leftover is $\boldsymbol{r}_{K,\Lambda,T}(h)$. $\square$

A.4 Orthogonal projection

Let $\Pi_{K,\Lambda,T}$ be the $L^2(P_T)$-orthogonal projection onto $\mathcal{N}_{K,\Lambda,T}$. In coordinates,

$$\Pi_{K,\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda})^{\dagger}\boldsymbol{Q}'_{K,\Lambda}.$$

The residual-maker is

$$\boldsymbol{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}. \tag{A.6}$$

It satisfies the projection identities

$$\boldsymbol{M}'_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}, \qquad \boldsymbol{M}^2_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}, \qquad \boldsymbol{Q}'_{K,\Lambda}\boldsymbol{M}_{K,\Lambda} = 0.$$

Lemma 2 (Projection identities for the residual-maker)**.** *Let $Q = \boldsymbol{Q}_{K,\Lambda}$, $G = Q'Q$,*

$$\Pi = QG^{\dagger}Q', \qquad M = I - \Pi,$$

where $G^{\dagger}$ is the Moore-Penrose inverse. Then Π and M are symmetric and idempotent, and M annihilates the sieve nuisance space:

$$M' = M, \qquad M^2 = M, \qquad MQ = 0, \qquad Q'M = 0.$$

Proof. Since $G = Q'Q$ is symmetric, its Moore-Penrose inverse $G^{\dagger}$ is also symmetric. Hence

$$\Pi' = \{QG^{\dagger}Q'\}' = QG^{\dagger}Q' = \Pi.$$

For idempotence, use the Moore-Penrose identity $G^{\dagger}GG^{\dagger} = G^{\dagger}$:

$$\Pi^2 = QG^{\dagger}Q'QG^{\dagger}Q' = QG^{\dagger}GG^{\dagger}Q' = QG^{\dagger}Q' = \Pi.$$

Therefore

$$M' = I - \Pi' = M, \qquad M^2 = (I - \Pi)^2 = I - 2\Pi + \Pi^2 = I - \Pi = M.$$

It remains to show that M kills the columns of Q. Since $\mathcal{R}(G) = \mathcal{R}(Q')$ and $G^{\dagger}G$ is the orthogonal projector onto $\mathcal{R}(G)$,

$$QG^{\dagger}G = Q.$$

Thus

$$\Pi Q = QG^{\dagger}Q'Q = QG^{\dagger}G = Q, \qquad MQ = (I - \Pi)Q = 0.$$

Finally, because M is symmetric, $Q'M = (MQ)' = 0$. This proves all projection identities. □

B Profiling Is Orthogonal Projection

For a trial value β, define the partial outcome

$$\boldsymbol{y}^{\circ}_{\beta} = \boldsymbol{y} - \beta\boldsymbol{x}.$$

It is the original outcome $\boldsymbol{y}$ after subtracting the trial linear component $\beta\boldsymbol{x}$.

Profiling the nuisance means finding the closest nuisance vector to the partial outcome $\boldsymbol{y}^\circ_\beta$ in $\mathcal{N}_{K,\Lambda,T}$:

$$\widehat{\boldsymbol{\mu}}_{\beta,\Lambda} = \arg\min_{\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}} \|\boldsymbol{y}^\circ_\beta - \boldsymbol{g}\|_T^2.$$

By Hilbert projection,

$$\widehat{\boldsymbol{\mu}}_{\beta,\Lambda} = \Pi_{K,\Lambda,T}\boldsymbol{y}^\circ_\beta, \qquad \widehat{\boldsymbol{u}}(\beta,\Lambda) = \boldsymbol{y}^\circ_\beta - \widehat{\boldsymbol{\mu}}_{\beta,\Lambda} = \boldsymbol{M}_{K,\Lambda}\boldsymbol{y}^\circ_\beta.$$

Lemma 3 (Profile normal equation)**.** *For every $\boldsymbol{h}\in\mathcal{N}_{K,\Lambda,T}$,*

$$\langle\widehat{\boldsymbol{u}}(\beta,\Lambda),\boldsymbol{h}\rangle_T = 0.$$

Equivalently,

$$\boldsymbol{Q}'_{K,\Lambda}\widehat{\boldsymbol{u}}(\beta,\Lambda) = 0.$$

Proof. This is the KKT condition for a least-squares projection in $L^2(P_T)$. If $\boldsymbol{h} = \boldsymbol{Q}_{K,\Lambda}\boldsymbol{a}$, then

$$\langle\widehat{\boldsymbol{u}},\boldsymbol{h}\rangle_T = T^{-1}\widehat{\boldsymbol{u}}'\boldsymbol{Q}_{K,\Lambda}\boldsymbol{a} = T^{-1}(\boldsymbol{Q}'_{K,\Lambda}\widehat{\boldsymbol{u}})'\boldsymbol{a} = 0.$$

□

B.1 Break search as subspace selection

For fixed β, the partial outcome is

$$\boldsymbol{y}^\circ_\beta = \boldsymbol{y} - \beta\boldsymbol{x}.$$

For a candidate partition $\Lambda = (k_1,\ldots,k_q)$, set $k_0 = 0$, $k_{q+1} = T$, and

$$I_j(\Lambda) = \{t : k_j < t \le k_{j+1}\}.$$

The block sieve matrix is

$$\boldsymbol{Q}_{K,\Lambda} = [\boldsymbol{D}_0(\Lambda)\boldsymbol{P},\ldots,\boldsymbol{D}_q(\Lambda)\boldsymbol{P}].$$

For $c = (c_0',\ldots,c_q')'$, the fitted nuisance vector is $\boldsymbol{Q}_{K,\Lambda}c$, with component

$$(\boldsymbol{Q}_{K,\Lambda}c)_t = \boldsymbol{P}_K(W_{t-1})'c_j \qquad \text{if } t\in I_j(\Lambda).$$

Thus minimizing over c is exactly the same as choosing a separate sieve coefficient vector c_j on each segment.

Lemma 4 (Profile RSS is a Hilbert distance)**.** *For fixed (β,Λ), define the segment cost*

$$C_K(s,e;\beta) = \min_{c\in\mathbb{R}^K}\sum_{t=s}^{e}\{y_t - \beta x_t - \boldsymbol{P}_K(W_{t-1})'c\}^2,$$

and the profile RSS

$$RSS_K(\beta,\Lambda)=\sum_{j=0}^{q} C_K(k_j+1,k_{j+1};\beta).$$

Then

$$\begin{aligned}RSS_K(\beta,\Lambda)&=\min_{c\in\mathbb{R}^{(q+1)K}}\|\boldsymbol{y}_\beta^\circ-\boldsymbol{Q}_{K,\Lambda}c\|_2^2\\&=\inf_{\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}}\|\boldsymbol{y}_\beta^\circ-\boldsymbol{g}\|_2^2\\&=T\,\mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ,\mathcal{N}_{K,\Lambda,T})\\&=T\|\boldsymbol{M}_{K,\Lambda}\boldsymbol{y}_\beta^\circ\|_T^2.\end{aligned}\tag{B.1}$$

Proof. Because each segment $I_j(\Lambda)$ has its own coefficient vector c_j,

$$\begin{aligned}RSS_K(\beta,\Lambda)&=\min_{c_0,\dots,c_q}\sum_{j=0}^{q}\sum_{t\in I_j(\Lambda)}\{y_t-\beta x_t-\boldsymbol{P}_K(W_{t-1})'c_j\}^2\\&=\min_{c\in\mathbb{R}^{(q+1)K}}\|\boldsymbol{y}-\beta\boldsymbol{x}-\boldsymbol{Q}_{K,\Lambda}c\|_2^2\\&=\min_{c\in\mathbb{R}^{(q+1)K}}\|\boldsymbol{y}_\beta^\circ-\boldsymbol{Q}_{K,\Lambda}c\|_2^2.\end{aligned}$$

Since $\mathcal{N}_{K,\Lambda,T}=\mathrm{col}(\boldsymbol{Q}_{K,\Lambda})$, every $\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}$ can be written as $\boldsymbol{g}=\boldsymbol{Q}_{K,\Lambda}c$. Therefore

$$\min_c\|\boldsymbol{y}_\beta^\circ-\boldsymbol{Q}_{K,\Lambda}c\|_2^2=\inf_{\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}}\|\boldsymbol{y}_\beta^\circ-\boldsymbol{g}\|_2^2.$$

The Euclidean norm and the empirical Hilbert norm satisfy $\|\boldsymbol{a}\|_2^2=T\|\boldsymbol{a}\|_T^2$. Hence

$$\inf_{\boldsymbol{g}\in\mathcal{N}_{K,\Lambda,T}}\|\boldsymbol{y}_\beta^\circ-\boldsymbol{g}\|_2^2=T\,\mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ,\mathcal{N}_{K,\Lambda,T}).$$

Finally, $\boldsymbol{M}_{K,\Lambda}$ is the residual-maker for the orthogonal projection onto $\mathcal{N}_{K,\Lambda,T}$, so

$$\mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ,\mathcal{N}_{K,\Lambda,T})=\|\boldsymbol{M}_{K,\Lambda}\boldsymbol{y}_\beta^\circ\|_T^2.$$

This proves (B.1). □

Consequently, for known order q, the break estimator is

$$\begin{aligned}\widehat{\Lambda}_q(\beta)&=\arg\min_{\Lambda\in\mathcal{L}_q(\epsilon)}RSS_K(\beta,\Lambda)\\&=\arg\min_{\Lambda\in\mathcal{L}_q(\epsilon)}\mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ,\mathcal{N}_{K,\Lambda,T}).\end{aligned}$$

Unknown order adds a penalty to the same distance:

$$\widehat{q}(\beta)=\arg\min_{0\le q\le\bar{q}}\left\{T\,\mathrm{dist}_T^2(\boldsymbol{y}_\beta^\circ,\mathcal{N}_{K,\widehat{\Lambda}_q(\beta),T})+\mathrm{pen}_T(q,K)\right\}.$$

Thus break search is subspace selection: each candidate Λ creates a candidate nuisance subspace $\mathcal{N}_{K,\Lambda,T}$, and RSS picks the subspace closest to the partial outcome $\boldsymbol{y}_\beta^\circ$.

C Residualized Score Direction

The raw score weight $\boldsymbol{w} \in L^2(P_T)$ may contain nuisance directions. The geometric score direction is its nuisance-orthogonal component

$$\boldsymbol{v}_\Lambda = \boldsymbol{w}^c(\Lambda) = \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}.$$

Then

$$\boldsymbol{v}_\Lambda \in \mathcal{N}_{K,\Lambda,T}^{\perp_T}, \qquad \langle \boldsymbol{v}_\Lambda, \boldsymbol{h}\rangle_T = 0 \quad \forall \boldsymbol{h} \in \mathcal{N}_{K,\Lambda,T}.$$

The sample profiled moment is the Hilbert inner product

$$\Psi_T(\beta, \Lambda) = \langle \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta\boldsymbol{x}), \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\rangle_T. \tag{C.1}$$

The scaled score is

$$S_T(\beta, \Lambda) = \sqrt{T}\,\Psi_T(\beta, \Lambda).$$

Lemma 5 (Finite-sample nuisance cancellation)**.** *If $\boldsymbol{g} \in \mathcal{N}_{K,\Lambda,T}$, then*

$$\langle \boldsymbol{M}_{K,\Lambda}\boldsymbol{g}, \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\rangle_T = 0.$$

Proof. Because $\boldsymbol{g} \in \mathcal{N}_{K,\Lambda,T}$, $\boldsymbol{M}_{K,\Lambda}\boldsymbol{g} = 0$. Hence the inner product is zero. Equivalently, $(\boldsymbol{Q}c)'\boldsymbol{M}_{K,\Lambda}\boldsymbol{w} = 0$ for every c. □

D Known True Partition: Wilks Logic in Geometry

Set $\Lambda = \Lambda_0$, and abbreviate

$$\boldsymbol{Q}_0 = \boldsymbol{Q}_{K,\Lambda_0}, \qquad \boldsymbol{M}_0 = \boldsymbol{M}_{K,\Lambda_0}, \qquad \boldsymbol{v}_0 = \boldsymbol{M}_0\boldsymbol{w}.$$

Decompose the true nuisance into its finite-sieve projection and approximation residual:

$$\boldsymbol{\mu}_0 = \boldsymbol{Q}_0\boldsymbol{c}_0 + \boldsymbol{r}_0, \qquad \boldsymbol{Q}_0\boldsymbol{c}_0 \in \mathcal{N}_{K,\Lambda_0,T}, \qquad \boldsymbol{r}_0 \in \mathcal{N}_{K,\Lambda_0,T}^{\perp_T}.$$

Lemma 6 (True-partition score decomposition)**.** *At $\beta = \beta_0$ and $\Lambda = \Lambda_0$, the projected score satisfies*

$$S_T(\beta_0, \Lambda_0) = T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0 + T^{-1/2}\boldsymbol{r}_0'\boldsymbol{v}_0. \tag{D.1}$$

The first term is the stochastic score. The second term is the sieve approximation-bias term, and this is the term controlled by approximation rates:

$$T^{-1/2}\boldsymbol{r}_0'\boldsymbol{v}_0 = T^{-1/2}\boldsymbol{r}_0'\boldsymbol{M}_0\boldsymbol{w} = o_p(1). \tag{D.2}$$

Proof. At the true parameter, the partial outcome is

$$\boldsymbol{y}^\circ_{\beta_0} = \boldsymbol{y} - \beta_0\boldsymbol{x} = \boldsymbol{\mu}_0 + \boldsymbol{u} = \boldsymbol{Q}_0\boldsymbol{c}_0 + \boldsymbol{r}_0 + \boldsymbol{u}.$$

Apply the true-partition residual-maker $\boldsymbol{M}_0$. Since $\boldsymbol{Q}_0\boldsymbol{c}_0$ is in $\mathcal{N}_{K,\Lambda_0,T}$, Lemma 2 gives $\boldsymbol{M}_0\boldsymbol{Q}_0\boldsymbol{c}_0 = 0$. Therefore

$$\boldsymbol{M}_0\boldsymbol{y}^\circ_{\beta_0} = \boldsymbol{M}_0\boldsymbol{u} + \boldsymbol{M}_0\boldsymbol{r}_0.$$

The projected score is

$$\begin{aligned} S_T(\beta_0, \Lambda_0) &= \sqrt{T}\langle \boldsymbol{M}_0 \boldsymbol{y}^{\circ}_{\beta_0}, \boldsymbol{M}_0 \boldsymbol{w}\rangle_T \\ &= T^{-1/2}(\boldsymbol{M}_0 \boldsymbol{u} + \boldsymbol{M}_0 \boldsymbol{r}_0)'\boldsymbol{v}_0. \end{aligned}$$

Because $\boldsymbol{M}_0$ is symmetric and idempotent and $\boldsymbol{v}_0 = \boldsymbol{M}_0\boldsymbol{w}$, we have $\boldsymbol{M}_0\boldsymbol{v}_0 = \boldsymbol{v}_0$. Hence

$$(\boldsymbol{M}_0\boldsymbol{u})'\boldsymbol{v}_0 = \boldsymbol{u}'\boldsymbol{v}_0, \qquad (\boldsymbol{M}_0\boldsymbol{r}_0)'\boldsymbol{v}_0 = \boldsymbol{r}_0'\boldsymbol{v}_0.$$

Substituting these two identities gives (D.1). The term $T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0$ is handled by the score CLT and variance consistency. The term $T^{-1/2}\boldsymbol{r}_0'\boldsymbol{v}_0$ is the only approximation-rate term in this decomposition, because it is the only term containing the sieve residual $\boldsymbol{r}_0$. □

E Omitted Breaks: False Predictivity as a Projection Angle

Let $\mathcal{N}_{K,0,T}$ be the stable/no-break nuisance space and let $\boldsymbol{M}_{\text{st}}$ be the residual-maker for orthogonal projection onto $\mathcal{N}_{K,0,T}$. Define

$$\boldsymbol{v}_{\text{st}} = \boldsymbol{M}_{\text{st}}\boldsymbol{w}, \qquad S_T^{\text{st}}(\beta) = \sqrt{T}\langle \boldsymbol{M}_{\text{st}}(\boldsymbol{y} - \beta\boldsymbol{x}), \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T.$$

Proposition 3 (Omitted-break projection drift)**.** *At the true parameter β_0,*

$$S_T^{\text{st}}(\beta_0) = T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_{\text{st}} + B_T, \qquad B_T = T^{-1/2}\boldsymbol{\mu}_0'\boldsymbol{v}_{\text{st}}. \tag{E.1}$$

Equivalently,

$$B_T = \sqrt{T}\langle \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0, \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T. \tag{E.2}$$

If

$$\langle \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0, \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T \to b \neq 0,$$

then $B_T/\sqrt{T} \to b$, so the stable score has deterministic drift of order $\sqrt{T}$ at the true parameter.

Proof. At β_0,

$$\boldsymbol{y} - \beta_0\boldsymbol{x} = \boldsymbol{\mu}_0 + \boldsymbol{u}.$$

Applying the stable residual-maker gives

$$\boldsymbol{M}_{\text{st}}(\boldsymbol{y} - \beta_0\boldsymbol{x}) = \boldsymbol{M}_{\text{st}}\boldsymbol{u} + \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0.$$

Therefore

$$\begin{aligned} S_T^{\text{st}}(\beta_0) &= T^{-1/2}(\boldsymbol{M}_{\text{st}}\boldsymbol{u} + \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0)'\boldsymbol{v}_{\text{st}} \\ &= T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_{\text{st}} + T^{-1/2}\boldsymbol{\mu}_0'\boldsymbol{v}_{\text{st}}, \end{aligned}$$

because $\boldsymbol{M}_{\text{st}}$ is symmetric and idempotent and $\boldsymbol{v}_{\text{st}} = \boldsymbol{M}_{\text{st}}\boldsymbol{w}$. This proves (E.1). The same symmetry and idempotence give

$$T^{-1/2}\boldsymbol{\mu}_0'\boldsymbol{v}_{\text{st}} = T^{-1/2}\boldsymbol{\mu}_0'\boldsymbol{M}_{\text{st}}\boldsymbol{w} = \sqrt{T}\langle \boldsymbol{M}_{\text{st}}\boldsymbol{\mu}_0, \boldsymbol{M}_{\text{st}}\boldsymbol{w}\rangle_T,$$

which proves (E.2). If the empirical inner product converges to $b \neq 0$, the drift is $B_T = \sqrt{T}\,b + o(\sqrt{T})$. □

The same statement can be read as an angle condition. On the stable normal space, define the empirical cosine

$$\cos_T(\boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0, \boldsymbol{v}_{\rm st}) = \frac{\langle \boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0, \boldsymbol{v}_{\rm st}\rangle_T}{\|\boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0\|_T \|\boldsymbol{v}_{\rm st}\|_T},$$

when both norms are nonzero. Then

$$B_T = \sqrt{T}\,\|\boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0\|_T\,\|\boldsymbol{v}_{\rm st}\|_T\,\cos_T(\boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0, \boldsymbol{v}_{\rm st}).$$

Thus false predictivity is a nonzero angle between the unremoved nuisance residue and the residualized score direction. The correct break-aware projection is needed because it makes the nuisance residue $\boldsymbol{M}_0\boldsymbol{\mu}_0$ small in the relevant normal space. As a toy example, suppose there are no covariates and the stable nuisance space contains only constants:

$$\mathcal{N}_{K,0,T} = \operatorname{span}\{\mathbf{1}_T\}.$$

Let the true nuisance contain one level shift,

$$\mu_{0,t} = \delta \mathbf{1}\{t > k_0\}, \qquad \pi_T = T^{-1}\sum_{t=1}^{T} \mathbf{1}\{t > k_0\}.$$

The stable projection removes only the global mean. Hence

$$(\boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0)_t = \delta\{\mathbf{1}(t > k_0) - \pi_T\}.$$

If the residualized score direction is aligned with this step, for instance $\boldsymbol{M}_{\rm st}\boldsymbol{w} = \boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0$, then

$$\langle \boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0, \boldsymbol{M}_{\rm st}\boldsymbol{w}\rangle_T = \|\boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0\|_T^2 = \delta^2\pi_T(1-\pi_T).$$

When $\delta \neq 0$ and $\pi_T \to \pi \in (0,1)$, the limit is $\delta^2\pi(1-\pi) \neq 0$. The stable score therefore reports a false signal even at β_0.

The correct break-aware block sieve changes the projection target. Under the true partition,

$$\boldsymbol{\mu}_0 = \boldsymbol{Q}_0\boldsymbol{c}_0 + \boldsymbol{r}_0, \qquad \boldsymbol{Q}_0\boldsymbol{c}_0 \in \mathcal{N}_{K,\Lambda_0,T}, \qquad \|\boldsymbol{r}_0\|_T = o_p(1),$$

so $\boldsymbol{M}_0\boldsymbol{\mu}_0 = \boldsymbol{r}_0$ is small. Under the stable space, $\boldsymbol{M}_{\rm st}\boldsymbol{\mu}_0$ need not be small. The block-sieve projection removes the low-frequency break component before testing β; otherwise omitted nuisance breaks can enter the score as false predictivity.

F Estimated Partition

The estimated break procedure chooses a random subspace

$$\widehat{\mathcal{N}}_T = \mathcal{N}_{K,\widehat{\Lambda},T}, \qquad \widehat{M}_T = \boldsymbol{M}_{K,\widehat{\Lambda}}.$$

The feasible score is

$$\widehat{S}_T(\beta_0) = \sqrt{T}\langle \widehat{M}_T(\boldsymbol{y} - \beta_0\boldsymbol{x}), \widehat{M}_T\boldsymbol{w}\rangle_T.$$

The true-partition score is

$$S_T^0(\beta_0) = \sqrt{T}\langle \boldsymbol{M}_0(\boldsymbol{y} - \beta_0 \boldsymbol{x}), \boldsymbol{M}_0 \boldsymbol{w}\rangle_T, \qquad \boldsymbol{M}_0 = \boldsymbol{M}_{K,\Lambda_0}.$$

Because each residual-maker is symmetric and idempotent,

$$\widehat{S}_T(\beta_0) = T^{-1/2}(\boldsymbol{\mu}_0 + \boldsymbol{u})'\widehat{M}_T \boldsymbol{w}, \qquad S_T^0(\beta_0) = T^{-1/2}(\boldsymbol{\mu}_0 + \boldsymbol{u})'\boldsymbol{M}_0 \boldsymbol{w}.$$

Therefore the exact estimated-partition score error is

$$\begin{aligned} \widehat{S}_T(\beta_0) - S_T^0(\beta_0) &= T^{-1/2}\boldsymbol{\mu}_0'(\widehat{M}_T - \boldsymbol{M}_0)\boldsymbol{w} \\ &\quad + T^{-1/2}\boldsymbol{u}'(\widehat{M}_T - \boldsymbol{M}_0)\boldsymbol{w}. \end{aligned} \tag{F.1}$$

The first term is local sieve-bias perturbation. The second term is local noise perturbation. Thus the estimated-partition Wilks argument needs

$$T^{-1/2}\boldsymbol{\mu}_0'(\widehat{M}_T - \boldsymbol{M}_0)\boldsymbol{w} = o_p(1), \qquad T^{-1/2}\boldsymbol{u}'(\widehat{M}_T - \boldsymbol{M}_0)\boldsymbol{w} = o_p(1). \tag{F.2}$$

For the variance, the corresponding requirement is

$$\widehat{V}_T - V_T^0 = o_p(1), \qquad \widehat{V}_T = P_T g_t(\beta_0, \widehat{\Lambda})^2, \qquad V_T^0 = P_T g_t(\beta_0, \Lambda_0)^2. \tag{F.3}$$

Equations (F.1)–(F.3) are the exact bridge from estimated breaks to the true-partition geometry.

The reason this is weaker than operator-norm convergence is visible from (F.1). We do not need $\|\widehat{M}_T - \boldsymbol{M}_0\|_{op} \to 0$. We only need the difference to be small when applied to the score direction $\boldsymbol{w}$ and paired with $\boldsymbol{\mu}_0$ and $\boldsymbol{u}$, plus the analogous second-moment replacement. In the directional topology of the geometric dictionary, this is a statement that

$$d_{\mathcal{D},T}(\widehat{M}_T, \boldsymbol{M}_0) = o_p(1)$$

for a direction set $\mathcal{D}_T$ containing the vectors that appear in the score, variance, and derivative. This is the topology induced by the theorem, not an after-the-fact weakening of an operator-norm claim.

The localization proof uses the RSS distance identity from Lemma 4. At β_0, set

$$\Delta_\Lambda = \boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0, \qquad \boldsymbol{y}^\circ_{\beta_0} = \boldsymbol{\mu}_0 + \boldsymbol{u}.$$

Then

$$\begin{aligned} RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0) &= (\boldsymbol{y}^\circ_{\beta_0})'\Delta_\Lambda \boldsymbol{y}^\circ_{\beta_0} \\ &= \boldsymbol{\mu}_0'\Delta_\Lambda \boldsymbol{\mu}_0 + 2\boldsymbol{\mu}_0'\Delta_\Lambda \boldsymbol{u} + \boldsymbol{u}'\Delta_\Lambda \boldsymbol{u}. \end{aligned} \tag{F.4}$$

The deterministic term is the geometric drift from choosing the wrong nuisance subspace. The two stochastic terms must be uniformly smaller than that drift on wrong partitions. Part II states the primitive concentration conditions and then uses (F.4) to obtain break localization and order selection.

G EL as a Finite-Sample Convex KKT Problem

For fixed (K, Λ, β), everything in this section is finite-sample algebra in $L^2(P_T)$. Define the profiled residual, score direction, and scalar moment array

$$\boldsymbol{r}_\beta = \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta \boldsymbol{x}), \qquad \boldsymbol{a} = \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}, \qquad g_t(\beta, \Lambda) = r_{\beta,t} a_t.$$

Then

$$\Psi_T(\beta, \Lambda) = P_T g(\beta, \Lambda) = T^{-1} \sum_{t=1}^{T} g_t(\beta, \Lambda) = \langle \boldsymbol{r}_\beta, \boldsymbol{a} \rangle_T.$$

Thus the sieve basis $Q_{K,\Lambda}$ has already represented the nuisance feasible directions: $\mathrm{col}(Q_{K,\Lambda})$ is the tangent plane for nuisance variation, and $M_{K,\Lambda}$ keeps only the orthogonal score component.

EL adds one more finite-dimensional convex problem. The primal feasible set is the probability simplex cut by the affine moment hyperplane

$$\mathcal{C}_T(\beta, \Lambda) = \left\{ \pi \in \mathbb{R}_+^T : \sum_{t=1}^{T} \pi_t = 1, \quad \sum_{t=1}^{T} \pi_t g_t(\beta, \Lambda) = 0 \right\}.$$

The empirical likelihood problem is equivalently

$$\min_{\pi \in \mathcal{C}_T(\beta, \Lambda)} - \sum_{t=1}^{T} \log(T\pi_t). \tag{G.1}$$

This is a convex barrier objective over a convex feasible set. Its Lagrangian is

$$\mathcal{L}(\pi, \eta, \lambda) = - \sum_{t=1}^{T} \log(T\pi_t) + \eta \left(\sum_{t=1}^{T} \pi_t - 1 \right) + T\lambda \sum_{t=1}^{T} \pi_t g_t.$$

The KKT first-order condition gives

$$-\frac{1}{\pi_t} + \eta + T\lambda g_t = 0.$$

After normalizing by $\sum_t \pi_t = 1$, the EL weights are

$$\widehat{\pi}_t(\lambda) = \frac{1}{T\{1 + \lambda g_t(\beta, \Lambda)\}}, \qquad 1 + \lambda g_t(\beta, \Lambda) > 0. \tag{G.2}$$

The remaining KKT equation is the dual moment equation

$$P_T \left[\frac{g_t(\beta, \Lambda)}{1 + \lambda g_t(\beta, \Lambda)} \right] = 0. \tag{G.3}$$

The log empirical likelihood ratio is therefore

$$\ell_T(\beta, \Lambda) = 2 \sum_{t=1}^{T} \log\{1 + \lambda_T g_t(\beta, \Lambda)\}. \tag{G.4}$$

The quadratic EL reduction is also mostly KKT algebra. If $\max_t |\lambda_T g_t| = o_p(1)$, then expanding (G.3) gives

$$0 = P_T\left[\frac{g_t}{1+\lambda_T g_t}\right] = P_T g_t - \lambda_T P_T g_t^2 + R_{1T}, \qquad R_{1T} = o_p(T^{-1/2}).$$

Hence, with $V_T = P_T g_t^2$,

$$\lambda_T = \frac{P_T g_t}{V_T} + o_p(T^{-1/2}).$$

Expanding (G.4) then yields

$$\ell_T(\beta_0, \Lambda) = \frac{T\{P_T g_t(\beta_0, \Lambda)\}^2}{V_T(\Lambda)} + o_p(1) = \frac{S_T(\beta_0, \Lambda)^2}{V_T(\Lambda)} + o_p(1). \tag{G.5}$$

The stochastic work is not to rediscover EL; it is only to verify the small size conditions under which the KKT expansion is valid.

H Population Hilbert Space

H.1 Population object in $L^2(P)$

To express breaks in population geometry, include rescaled time $S \in (0, 1]$. Let

$$O = (Y, X, W, S, U)$$

be a population random element with law P. The true regime map is

$$j_0(S) = j \quad \Longleftrightarrow \quad \tau_{j,0} < S \le \tau_{j+1,0}.$$

The population model is the identity in $L^2(P)$:

$$Y = \beta_0 X + \mu_0(S, W) + U, \qquad \mu_0(S, W) = \sum_{j=0}^{q_0} m_j(W)\mathbf{1}\{\tau_{j,0} < S \le \tau_{j+1,0}\}. \tag{H.1}$$

H.2 Population nuisance space

For a population partition Λ, define

$$\mathcal{N}_\Lambda^P = \overline{\left\{\sum_{j=0}^{q} \mathbf{1}\{S \in I_j(\Lambda)\}h_j(W) : h_j \in L^2(P_W)\right\}}^{L^2(P)}.$$

Let Π_Λ^P be the $L^2(P)$-orthogonal projection onto $\mathcal{N}_\Lambda^P$, and let

$$M_\Lambda^P = I - \Pi_\Lambda^P.$$

For $\beta \in \mathcal{B}$, define the population partial outcome

$$Y_\beta^\circ = Y - \beta X.$$

The population profiled residual and score direction are

$$r_\beta^P(\Lambda) = M_\Lambda^P Y_\beta^\circ = M_\Lambda^P (Y - \beta X), \qquad v^P(\Lambda) = M_\Lambda^P w.$$

The population moment is

$$\Psi(\beta, \Lambda) = \langle r_\beta^P(\Lambda), v^P(\Lambda) \rangle_P. \tag{H.2}$$

I Population Identification

At the true population partition Λ_0, abbreviate

$$\mathcal{N}_0^P = \mathcal{N}_{\Lambda_0}^P, \qquad \Pi_0^P = \Pi_{\Lambda_0}^P, \qquad M_0^P = M_{\Lambda_0}^P, \qquad v_0^P = v^P(\Lambda_0) = M_0^P w.$$

Because the true nuisance is regime-specific with the true population partition,

$$\mu_0(S, W) \in \mathcal{N}_0^P.$$

Therefore the true-partition population residual-maker removes it:

$$M_0^P \mu_0 = 0. \tag{I.1}$$

This is the $L^2(P)$ analog of the finite-sample statement that the correct break-aware nuisance space absorbs the nuisance component.

Using the population model (H.1), the population partial outcome at a generic β is

$$\begin{aligned} Y_\beta^\circ &= Y - \beta X \\ &= \mu_0(S, W) + U - (\beta - \beta_0) X. \end{aligned}$$

Projecting onto $(\mathcal{N}_0^P)^{\perp_P}$ gives

$$\begin{aligned} r_\beta^P(\Lambda_0) &= M_0^P Y_\beta^\circ \\ &= M_0^P \mu_0 + M_0^P U - (\beta - \beta_0) M_0^P X \\ &= M_0^P U - (\beta - \beta_0) M_0^P X, \end{aligned} \tag{I.2}$$

because $M_0^P \mu_0 = 0$. Taking the inner product with the true population score direction v_0^P gives

$$\Psi(\beta, \Lambda_0) = \langle M_0^P U, v_0^P \rangle_P - (\beta - \beta_0) \langle M_0^P X, v_0^P \rangle_P. \tag{I.3}$$

This derivation is algebraic. The two population assumptions enter only after (I.3) is obtained.

The exogeneity condition is

$$\langle M_0^P U, v_0^P \rangle_P = 0.$$

The relevance condition is

$$\Gamma = \langle M_0^P X, v_0^P \rangle_P \neq 0.$$

Geometrically this is a transversality condition. After quotienting out the nuisance tangent space, the parametric direction is not invisible to the chosen normal score direction:

$$M_0^P X \not\perp_P M_0^P w.$$

Equivalently, the β-line crosses the population moment hyperplane non-tangentially. If $M_0^P X$ were orthogonal to every usable residualized score direction, the parametric direction would be swallowed by the nuisance space and β would not be identified by this moment.

Under these two conditions,

$$\Psi(\beta, \Lambda_0) = -(\beta - \beta_0)\Gamma.$$

Thus

$$\Psi(\beta, \Lambda_0) = 0 \quad \Longleftrightarrow \quad \beta = \beta_0. \tag{I.4}$$

Part III

Small Stochastic Bounds

J Small Stochastic Bounds

Part I has already fixed the objects. For every candidate partition Λ, write

$$\boldsymbol{r}_\beta(\Lambda) = \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta\boldsymbol{x}), \qquad \boldsymbol{a}(\Lambda) = \boldsymbol{M}_{K,\Lambda}\boldsymbol{w},$$

and

$$g_t(\beta, \Lambda) = r_{\beta,t}(\Lambda)a_t(\Lambda), \qquad \Psi_T(\beta, \Lambda) = P_T g(\beta, \Lambda) = \langle \boldsymbol{r}_\beta(\Lambda), \boldsymbol{a}(\Lambda)\rangle_T.$$

At the true partition, abbreviate

$$\boldsymbol{M}_0 = \boldsymbol{M}_{K,\Lambda_0}, \qquad \boldsymbol{v}_0 = \boldsymbol{M}_0\boldsymbol{w}, \qquad \boldsymbol{y}_\beta^\circ = \boldsymbol{y} - \beta\boldsymbol{x}.$$

For the estimated partition, write

$$\widehat{\boldsymbol{M}} = \boldsymbol{M}_{K,\widehat{\Lambda}}, \qquad \widehat{\boldsymbol{v}} = \widehat{\boldsymbol{M}}\boldsymbol{w}.$$

The variance objects are

$$V_T(\Lambda) = P_T g_t(\beta_0, \Lambda)^2, \qquad V_T^0 = V_T(\Lambda_0), \qquad \widehat{V}_T = V_T(\widehat{\Lambda}).$$

The stochastic part proves the following targets one by one:

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \widehat{\Lambda}) - \Psi(\beta, \Lambda_0)| = o_p(1), \tag{J.1}$$

$$\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda}) \Rightarrow N(0, V), \qquad \widehat{V}_T \to_p V > 0,$$

and the local empirical-likelihood conditions that make the KKT expansion (G.5) valid. None of these statements changes the geometric objects. They only control their sampling error.

J.1 Standing stochastic conditions

Let $\mathcal{L}_T$ denote the admissible set of candidate partitions searched by the estimator. For local break-date arguments, let $\mathcal{N}_T(r_T) \subset \mathcal{L}_T$ denote the event/set of partitions whose break dates are within the localization radius r_T of Λ_0. The following assumptions are the stochastic input for Part II. They are stated in the workbook notation so each proof below can be read directly.

Assumption 2 (Empirical-to-population inner-product bridge)**.** *Uniformly over $\Lambda \in \mathcal{L}_T$, the empirical Gram matrices of the block sieve design are stable:*

$$\sup_{\Lambda \in \mathcal{L}_T} \left\| T^{-1} \boldsymbol{Q}'_{K,\Lambda} \boldsymbol{Q}_{K,\Lambda} - G_Q(\Lambda) \right\|_{op} = o_p(1),$$

where

$$0 < c \leq \lambda_{\min}\{G_Q(\Lambda)\} \leq \lambda_{\max}\{G_Q(\Lambda)\} \leq C < \infty \qquad \textit{uniformly over } \Lambda \in \mathcal{L}_T.$$

The same uniform empirical laws hold for the projected cross moments involving $\boldsymbol{Q}_{K,\Lambda}$, $\boldsymbol{x}$, $\boldsymbol{w}$, and $\boldsymbol{u}$. Equivalently, for the finite sieve and break classes used here,

$$\sup_{f,g \in \mathcal{F}_{K,T}} |\langle \boldsymbol{f}_T, \boldsymbol{g}_T \rangle_T - \langle f, g \rangle_P| = o_p(1).$$

This is the bridge between $L^2(P_T)$ and $L^2(P)$. The spaces are not being identified as the same space; only the relevant inner products converge.

Assumption 3 (Score-negligible sieve remainder)**.** *For the true nuisance vector $\boldsymbol{\mu}_0$, the true-partition finite-sieve remainder*

$$\boldsymbol{r}_{\mu,T} = \boldsymbol{M}_0 \boldsymbol{\mu}_0$$

satisfies

$$\|\boldsymbol{r}_{\mu,T}\|_T = o_p(1), \qquad \sqrt{T}\, |\langle \boldsymbol{r}_{\mu,T}, \boldsymbol{v}_0 \rangle_T| = o_p(1), \qquad P_T\{r^2_{\mu,T,t} v^2_{0,t}\} = o_p(1).$$

The first bound is the consistency-scale approximation bound. The second is the score-scale bias bound. The third removes the sieve remainder from the variance.

Assumption 4 (Estimated-partition primitive closure)**.** *The RSS break estimator localizes at radius r_T:*

$$\Pr\{\widehat{\Lambda} \in \mathcal{N}_T(r_T)\} \to 1.$$

On $\mathcal{N}_T(r_T)$, local perturbations of the projected score and second moment are negligible:

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} \sqrt{T}\, |\Psi_T(\beta_0, \Lambda) - \Psi_T(\beta_0, \Lambda_0)| = o_p(1),$$

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} |V_T(\Lambda) - V_T^0| = o_p(1),$$

and, at the unscaled consistency level,

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} \sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \Lambda) - \Psi_T(\beta, \Lambda_0)| = o_p(1).$$

A sufficient local perturbation route is

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} T^{-1/2} |\boldsymbol{r}'_{\mu,T}(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}| = o_p(1),$$

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} T^{-1/2} |\boldsymbol{u}'(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}| = o_p(1),$$

with the variance replacement below. For a one-break local window, a standard sufficient rate is

$$\zeta_K \sqrt{\frac{K r_T}{T}} \to 0,$$

where ζ_K controls the sieve-envelope size.

Assumption 5 (RSS concentration and penalty rates)**.** *Let $\Delta_\Lambda = \boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0$. Outside the local set $\mathcal{N}_T(r_T)$, the deterministic RSS drift dominates stochastic RSS fluctuation:*

$$\inf_{\Lambda \in \mathcal{L}_T \setminus \mathcal{N}_T(r_T)} T^{-1} \boldsymbol{\mu}'_0 \Delta_\Lambda \boldsymbol{\mu}_0 \geq c_T,$$

$$\sup_{\Lambda \in \mathcal{L}_T \setminus \mathcal{N}_T(r_T)} T^{-1} |2\boldsymbol{\mu}'_0 \Delta_\Lambda \boldsymbol{u} + \boldsymbol{u}' \Delta_\Lambda \boldsymbol{u}| = o_p(c_T).$$

One primitive route for the stochastic bound is conditionally sub-Gaussian innovations plus sieve entropy/concentration over the searched RSS class.

For unknown order, the underfitted deterministic loss and overfitted stochastic gain satisfy

$$\inf_{q < q_0} \{RSS_K(q) - RSS_K(q_0)\} \geq cT\Delta_T^2 + o_p(T\Delta_T^2),$$

$$\sup_{q > q_0} \{RSS_K(q_0) - RSS_K(q)\} = O_p(K \log T),$$

and the penalty obeys

$$K \log T = o\{\mathrm{pen}_T(q, K)\}, \qquad \mathrm{pen}_T(q, K) = o(T\Delta_T^2)$$

for every fixed nonzero order difference.

Assumption 6 (Projected LLN and CLT at the true partition)**.** *The true-partition projected empirical moments obey*

$$\langle \boldsymbol{M}_0 \boldsymbol{u}, \boldsymbol{v}_0 \rangle_T - \langle M_0^P U, v_0^P \rangle_P = o_p(1),$$

$$\langle \boldsymbol{M}_0 \boldsymbol{x}, \boldsymbol{v}_0 \rangle_T - \langle M_0^P X, v_0^P \rangle_P = o_p(1),$$

and

$$T^{-1/2} \boldsymbol{u}' \boldsymbol{v}_0 \Rightarrow N(0, V), \qquad 0 < V < \infty.$$

Assumption 7 (Second and higher moment control)**.** *With* $\tilde{g}_t^0 = (\boldsymbol{M}_0\boldsymbol{u})_t v_{0,t}$,

$$P_T(\tilde{g}_t^0)^2 \to_p V, \qquad P_T|g_t(\beta_0, \Lambda_0)|^3 = O_p(1), \qquad P_T|g_t(\beta_0, \widehat{\Lambda})|^3 = O_p(1),$$

and

$$\max_{t \le T} |g_t(\beta_0, \Lambda_0)|/\sqrt{T} = o_p(1), \qquad \max_{t \le T} |g_t(\beta_0, \widehat{\Lambda})|/\sqrt{T} = o_p(1).$$

Assumption 8 (Scalar convex-hull condition)**.** *With probability tending to one, both scalar samples* $\{g_t(\beta_0, \Lambda_0) : 1 \le t \le T\}$ *and* $\{g_t(\beta_0, \widehat{\Lambda}) : 1 \le t \le T\}$ *contain at least one strictly positive value and at least one strictly negative value.*

J.2 Primitive stochastic assumptions sufficient for Part II

The high-level assumptions above are implied by familiar primitive bounds. Let

$$v_{t,0}^* = v_{0,t} = (\boldsymbol{M}_0\boldsymbol{w})_t, \qquad \widehat{v}_{t,0}^* = (\widehat{\boldsymbol{M}}\boldsymbol{w})_t.$$

A useful sufficient primitive package is:

1. **Envelope control.**

$$\max_{t \le T} |v_{t,0}^*| = O_p(1 + \zeta_K), \qquad \max_{t \le T} |\widehat{v}_{t,0}^*| = O_p(1 + \zeta_K).$$

2. **Variance replacement.** If $\sigma_t^2 = \mathbb{E}(u_t^2 \mid \mathcal{F}_{t-1})$, then

$$T^{-1} \sum_{t=1}^{T} \sigma_t^2 \{(\widehat{v}_{t,0}^*)^2 - (v_{t,0}^*)^2\} = o_p(1).$$

3. **Local sieve-bias stability.**

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} T^{-1/2} |\boldsymbol{r}'_{\mu,T}(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}| = o_p(1).$$

4. **Local noise perturbation.**

$$\sup_{\Lambda \in \mathcal{N}_T(r_T)} T^{-1/2} |\boldsymbol{u}'(\boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0)\boldsymbol{w}| = o_p(1),$$

   for example under the rate $\zeta_K\sqrt{K r_T / T} \to 0$ in the one-break local perturbation calculation.

5. **RSS concentration.** The errors have enough tail control, for example conditional sub-Gaussian tails, so the searched RSS class satisfies the concentration bounds in Assumption 5.

6. **Penalty separation.** The unknown-order penalty dominates searched overfit gains but is smaller than the underfit deterministic loss:

$$K \log T = o\{\mathrm{pen}_T(q, K)\}, \qquad \mathrm{pen}_T(q, K) = o(T\Delta_T^2).$$

These primitive conditions are not new geometry. They are the stochastic closure that makes the geometric KKT construction usable with estimated breaks.

J.3 Part II primitive proof-claim closure

This subsection records the detailed primitive route behind the high-level Part II assumptions. It is not a second model. It is the proof claim that the RSS localization, projected score CLT, variance replacement, max-score bound, and EL KKT expansion can all be obtained from standard local projection and concentration bounds.

Write

$$d_K = (q_0 + 1)K, \qquad b_K = 1 + \zeta_K, \qquad \mathbf{\Pi}_0 = I - \boldsymbol{M}_0.$$

At the true partition let $\boldsymbol{Q}_0 = \boldsymbol{Q}_{K,\Lambda_0}$. Assume the true nuisance has the finite-sieve decomposition

$$\boldsymbol{\mu}_0 = \boldsymbol{Q}_0 c_0 + \boldsymbol{r}_{\mu,T}.$$

Lemma 7 (Scalar EL KKT proof claim)**.** *Let $Z_{t,T}$ be any scalar triangular moment array, and define*

$$A_T = T^{-1/2} \sum_{t=1}^{T} Z_{t,T}, \qquad B_T = T^{-1} \sum_{t=1}^{T} Z_{t,T}^2, \qquad m_T = \max_{t \le T} |Z_{t,T}|.$$

If $A_T = O_p(1)$, $B_T \ge c + o_p(1)$, $m_T = o_p(\sqrt{T})$, and the scalar convex hull contains zero in its interior with probability tending to one, then the EL dual root λ_T satisfies

$$\lambda_T = \frac{T^{-1} \sum_t Z_{t,T}}{T^{-1} \sum_t Z_{t,T}^2} + o_p(T^{-1/2}),$$

and

$$2 \sum_{t=1}^{T} \log(1 + \lambda_T Z_{t,T}) = \frac{A_T^2}{B_T} + o_p(1).$$

Proof. The KKT equation is

$$0 = \sum_t \frac{Z_{t,T}}{1 + \lambda_T Z_{t,T}}, \qquad 1 + \lambda_T Z_{t,T} > 0.$$

With $S_1 = \sum_t Z_{t,T}$ and $S_2 = \sum_t Z_{t,T}^2$, the exact identity

$$S_1 = \lambda_T \sum_t \frac{Z_{t,T}^2}{1 + \lambda_T Z_{t,T}}$$

implies $|\lambda_T| = O_p(T^{-1/2})$ because $S_1 = O_p(\sqrt{T})$, $S_2 \ge cT + o_p(T)$, and $|S_1| m_T = o_p(T)$. Hence $|\lambda_T| m_T = o_p(1)$. Expanding the KKT equation around zero gives

$$0 = S_1 - \lambda_T S_2 + o_p(\sqrt{T}), \qquad \lambda_T = S_1/S_2 + o_p(T^{-1/2}).$$

Taylor expansion of the log ratio gives

$$2 \sum_t \log(1 + \lambda_T Z_{t,T}) = 2\lambda_T S_1 - \lambda_T^2 S_2 + o_p(1) = S_1^2/S_2 + o_p(1).$$

Since $S_1^2/S_2 = A_T^2/B_T$, the claim follows. $\square$

Lemma 8 (True-partition feasible-to-oracle proof claim)**.** *At* (β_0, Λ_0)*,*

$$\sqrt{T}\Psi_T(\beta_0, \Lambda_0) = T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0 + T^{-1/2}\boldsymbol{r}'_{\mu,T}\boldsymbol{v}_0.$$

If $T^{-1/2}\boldsymbol{r}'_{\mu,T}\boldsymbol{v}_0 = o_p(1)$*, then the score numerator is asymptotically the oracle numerator* $T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0$*. Moreover, if*

$$\|(\boldsymbol{M}_0\boldsymbol{r}_{\mu,T} - \boldsymbol{\Pi}_0\boldsymbol{u}) \odot \boldsymbol{v}_0\|_T = o_p(1), \qquad \|(\boldsymbol{M}_0\boldsymbol{r}_{\mu,T} - \boldsymbol{\Pi}_0\boldsymbol{u}) \odot \boldsymbol{v}_0\|_\infty = o_p(\sqrt{T}),$$

then

$$V_T^0 = T^{-1}\sum_{t=1}^{T} u_t^2 v_{0,t}^2 + o_p(1), \qquad \max_t |g_t(\beta_0, \Lambda_0)| = \max_t |u_t v_{0,t}| + o_p(\sqrt{T}).$$

Proof. Since $\boldsymbol{M}_0\boldsymbol{Q}_0 = 0$,

$$\boldsymbol{M}_0(\boldsymbol{y} - \beta_0\boldsymbol{x}) = \boldsymbol{M}_0(\boldsymbol{\mu}_0 + \boldsymbol{u}) = \boldsymbol{r}_{\mu,T} + \boldsymbol{M}_0\boldsymbol{u}.$$

Because $\boldsymbol{v}_0 = \boldsymbol{M}_0\boldsymbol{w}$ and $\boldsymbol{M}_0$ is symmetric and idempotent,

$$(\boldsymbol{M}_0\boldsymbol{u})'\boldsymbol{v}_0 = \boldsymbol{u}'\boldsymbol{v}_0.$$

This proves the numerator identity.

For the variance and max bounds, write

$$\boldsymbol{M}_0\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{\Pi}_0\boldsymbol{u}$$

and therefore

$$g_t(\beta_0, \Lambda_0) = u_t v_{0,t} + d_{t,0}, \qquad \boldsymbol{d}_0 = (\boldsymbol{M}_0\boldsymbol{r}_{\mu,T} - \boldsymbol{\Pi}_0\boldsymbol{u}) \odot \boldsymbol{v}_0.$$

The $L^2(P_T)$ bound on $\boldsymbol{d}_0$ gives the second-moment replacement by Cauchy-Schwarz, using $T^{-1}\sum_t u_t^2 v_{0,t}^2 = O_p(1)$. The sup-norm bound gives the max-score replacement. □

The preceding display is the place where primitive projection rates enter. A standard sufficient package is

$$\|\boldsymbol{M}_0\boldsymbol{r}_{\mu,T}\|_T = O_p(a_K), \qquad \|\boldsymbol{\Pi}_0\boldsymbol{u}\|_T = O_p\left(\sqrt{d_K/T}\right),$$

$$\|\boldsymbol{M}_0\boldsymbol{r}_{\mu,T}\|_\infty = O_p(a_K b_K), \qquad \|\boldsymbol{\Pi}_0\boldsymbol{u}\|_\infty = O_p\left(\zeta_K\sqrt{d_K/T}\right),$$

together with $\|\boldsymbol{v}_0\|_\infty = O_p(b_K)$. These imply

$$\|\boldsymbol{d}_0\|_T = O_p\left(a_K b_K + b_K\sqrt{d_K/T}\right) = o_p(1),$$

and

$$\|\boldsymbol{d}_0\|_\infty = O_p\left(a_K b_K^2 + \zeta_K b_K\sqrt{d_K/T}\right) = o_p(\sqrt{T}).$$

Lemma 9 (Oracle score proof claim)**.** *Suppose either $\boldsymbol{v}_0$ is admissible for the martingale array, or there is a predictable approximation $v_{t,0}^{\mathrm{pr}}$ satisfying*

$$T^{-1/2}\sum_t u_t(v_{0,t}-v_{t,0}^{\mathrm{pr}})=o_p(1),$$

$$T^{-1}\sum_t u_t^2\{v_{0,t}^2-(v_{t,0}^{\mathrm{pr}})^2\}=o_p(1), \qquad \max_t |v_{t,0}^{\mathrm{pr}}|=O_p(b_K).$$

If $b_K^{2+\delta_0}T^{-\delta_0/2}\to 0$ for some $\delta_0>0$, then

$$T^{-1/2}\sum_t u_t v_{0,t}\Rightarrow N(0,V),$$

$$T^{-1}\sum_t u_t^2 v_{0,t}^2\to_p V, \qquad \max_t |u_t v_{0,t}|=o_p(\sqrt{T}).$$

Proof. Apply the martingale triangular-array CLT to $\xi_{t,T}=T^{-1/2}u_t v_{t,0}^{\mathrm{pr}}$. The conditional variance is

$$T^{-1}\sum_t \sigma_t^2(v_{t,0}^{\mathrm{pr}})^2\to_p V,$$

and the conditional Lyapunov term is bounded by

$$CT^{-(1+\delta_0/2)}\sum_t |v_{t,0}^{\mathrm{pr}}|^{2+\delta_0}\le Cb_K^{2+\delta_0}T^{-\delta_0/2}=o_p(1).$$

The predictable-approximation assumption transfers the CLT back to $\boldsymbol{v}_0$. The variance LLN follows by applying the same martingale moment argument to $(u_t^2-\sigma_t^2)(v_{t,0}^{\mathrm{pr}})^2$, and the max bound follows from

$$\mathbb{P}\{\max_t |u_t v_{t,0}^{\mathrm{pr}}|>\varepsilon\sqrt{T}\}\le C_\varepsilon b_K^{2+\delta_0}T^{-\delta_0/2}+o(1).$$

□

Lemma 10 (Local estimated-partition bridge proof claim)**.** *For one break, let $\Lambda_0=(k_0)$, $\widetilde{\Lambda}=(\widetilde{k})$, and $h=|\widetilde{k}-k_0|=O_p(r_T)$. Under local Gram stability,*

$$T^{-1/2}\boldsymbol{\mu}_0'(\boldsymbol{M}_{K,\widetilde{k}}-\boldsymbol{M}_0)\boldsymbol{w}=O_p\left(\frac{b_K r_T}{\sqrt{T}}\right)+o_p(1),$$

and

$$T^{-1/2}\boldsymbol{u}'(\boldsymbol{M}_{K,\widetilde{k}}-\boldsymbol{M}_0)\boldsymbol{w}=O_p\left(\zeta_K\sqrt{\frac{Kr_T}{T}}\right)+o_p(1).$$

If

$$\frac{b_K r_T}{\sqrt{T}}\to 0, \qquad \zeta_K\sqrt{\frac{Kr_T}{T}}\to 0,$$

then

$$\sqrt{T}\{\Psi_T(\beta_0,\widetilde{\Lambda})-\Psi_T(\beta_0,\Lambda_0)\}=o_p(1).$$

If also $\zeta_K^2 Kr_T/T\to 0$, then

$$V_T(\widetilde{\Lambda})-V_T(\Lambda_0)=o_p(1).$$

Proof. Take $\widetilde{k} = k_0 + h$; the other side is symmetric. The only observations whose regime label changes are in $H = \{k_0 + 1, \dots, k_0 + h\}$. Relative to the candidate partition, the true nuisance can be written as

$$\boldsymbol{\mu}_0 = \boldsymbol{Q}_{K,\widetilde{k}} c^{(\widetilde{k})} + \boldsymbol{d}_H + \boldsymbol{r}_{\mu,T},$$

where $\boldsymbol{d}_H$ is supported on H and $\|\boldsymbol{d}_H\|_\infty = O_p(1)$. Since $\boldsymbol{M}_{K,\widetilde{k}}\boldsymbol{Q}_{K,\widetilde{k}} = 0$ and $\boldsymbol{M}_0\boldsymbol{Q}_0 = 0$,

$$\boldsymbol{\mu}_0'(\boldsymbol{M}_{K,\widetilde{k}} - \boldsymbol{M}_0)\boldsymbol{w} = \boldsymbol{d}_H'\boldsymbol{M}_{K,\widetilde{k}}\boldsymbol{w} + \boldsymbol{r}_{\mu,T}'(\boldsymbol{M}_{K,\widetilde{k}} - \boldsymbol{M}_0)\boldsymbol{w}.$$

The first term is bounded by $\|\boldsymbol{d}_H\|_1\|\boldsymbol{M}_{K,\widetilde{k}}\boldsymbol{w}\|_\infty = O_p(hb_K)$, and the second term is the local sieve-bias stability term. This proves the nuisance perturbation bound.

For the stochastic perturbation, write the residualized weights on the three blocks $A = \{1, \dots, k_0\}$, H, and $B = \{k_0 + h + 1, \dots, T\}$. The least-squares coefficient update on a long segment has the form

$$\widehat{\gamma}_{A\cup H} - \widehat{\gamma}_A = (P_{A\cup H}'P_{A\cup H})^{-1}P_H'(w_H - P_H\widehat{\gamma}_A),$$

so local Gram stability gives

$$\|\widehat{\gamma}_{A\cup H} - \widehat{\gamma}_A\| = O_p(h\zeta_K b_K/T),$$

and similarly on the right side. After cancellation, the remaining stochastic terms are controlled by

$$\|P_A'u_A\| + \|P_B'u_B\| = O_p(\sqrt{TK}), \qquad \|P_H'u_H\| = O_p(\zeta_K\sqrt{h}).$$

Multiplying by $T^{-1/2}$ gives the displayed $O_p\{\zeta_K\sqrt{Kr_T/T}\}$ bound. The denominator replacement follows by the same local boundary argument: $v_{\widetilde{k}} - v_{k_0}$ is order $O_p(b_K)$ only on the h-point boundary and order $O_p(h\zeta_K^2 b_K/T)$ away from it, so

$$|V_T(\widetilde{\Lambda}) - V_T(\Lambda_0)| = O_p\left(\frac{b_K^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1).$$

The stated rates make this $o_p(1)$. A fixed finite number of breaks is handled by summing the same boundary calculation over the finitely many local neighborhoods. □

Proposition 4 (Localization and order-selection proof claim)**.** *In the one-break calculation, suppose*

$$\Delta RSS_K(k) = RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0)$$

satisfies, uniformly for $h = |k - k_0|$,

$$E\{\Delta RSS_K(k) \mid D_T\} \geq ch\Delta_T^2 - o_p(h\Delta_T^2),$$

and

$$|\Delta RSS_K(k) - E\{\Delta RSS_K(k) \mid D_T\}| = O_p(\sqrt{hK\log T}) + O_p(K\log T).$$

Then with $R_T = K\log T$ and

$$r_T = R_T \max\{\Delta_T^{-2}, \Delta_T^{-4}\},$$

the known-order break estimator satisfies

$$|\widehat{k} - k_0| = O_p(r_T).$$

If Δ_T is bounded away from zero, this reduces to $|\widehat{k} - k_0| = O_p(K \log T)$. With fixed $\bar{q}$, if the overfit penalty increment dominates $K \log T$ and the underfit penalty saving is $o(T\Delta_T^2)$, then $\mathbb{P}(\widehat{q} = q_0) \to 1$.

Proof. The RSS identity from the geometric part gives

$$\Delta RSS_K(k) = \boldsymbol{\mu}_0' A_k \boldsymbol{\mu}_0 + 2\boldsymbol{\mu}_0' A_k \boldsymbol{u} + \boldsymbol{u}' A_k \boldsymbol{u}, \qquad A_k = \boldsymbol{M}_{K,k} - \boldsymbol{M}_{K,k_0}.$$

Conditional centering and equal-rank residual-makers give $E(\boldsymbol{u}' A_k \boldsymbol{u} \mid D_T) = \sigma^2 \operatorname{tr}(A_k) = 0$, so the conditional mean is the deterministic drift. The displayed stochastic fluctuation bound comes from a sub-Gaussian bound for the linear term and a Hanson-Wright bound for the rank-$O(K)$ quadratic term, followed by a union bound over searched break dates.

For $h \geq M r_T$,

$$\frac{R_T}{h\Delta_T^2} \leq M^{-1}, \qquad \frac{\sqrt{hR_T}}{h\Delta_T^2} \leq M^{-1/2}.$$

Choosing M large makes the stochastic part smaller than the positive drift, so $\Delta RSS_K(k) > 0$ uniformly outside the $M r_T$ window with probability tending to one. Since $\widehat{k}$ minimizes RSS and k_0 is admissible, $|\widehat{k} - k_0| = O_p(r_T)$.

For unknown order, underfitting loses at least $cT\Delta_T^2 + o_p(T\Delta_T^2)$ in deterministic RSS, while the penalty saving is $o(T\Delta_T^2)$. Overfitting can improve RSS only by $O_p(K \log T)$, while the penalty increase is larger than $K \log T$. A finite union over $q \leq \bar{q}$ gives $\mathbb{P}(\widehat{q} = q_0) \to 1$. □

J.4 RSS localization and order selection

Proposition 5 (RSS localization)**.** *Under Assumption 5, the known-order RSS estimator satisfies*

$$\Pr\{\widehat{\Lambda} \in \mathcal{N}_T(r_T)\} \to 1.$$

Proof. For any candidate Λ, Section 7 gives the exact decomposition

$$RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0) = \boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{\mu}_0 + 2\boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{u} + \boldsymbol{u}' \Delta_\Lambda \boldsymbol{u},$$

where $\Delta_\Lambda = \boldsymbol{M}_{K,\Lambda} - \boldsymbol{M}_0$. On $\mathcal{L}_T \setminus \mathcal{N}_T(r_T)$, Assumption 5 gives

$$T^{-1} \boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{\mu}_0 \geq c_T$$

uniformly, while

$$T^{-1} |2\boldsymbol{\mu}_0' \Delta_\Lambda \boldsymbol{u} + \boldsymbol{u}' \Delta_\Lambda \boldsymbol{u}| = o_p(c_T)$$

uniformly. Therefore

$$\inf_{\Lambda \in \mathcal{L}_T \setminus \mathcal{N}_T(r_T)} \{RSS_K(\beta_0, \Lambda) - RSS_K(\beta_0, \Lambda_0)\} > 0$$

with probability tending to one. A minimizer over the known-order search set cannot lie outside $\mathcal{N}_T(r_T)$ on this event. □

Proposition 6 (Unknown-order penalty selection)**.** *Under Assumption 5, the penalized RSS selector satisfies*

$$\Pr(\widehat{q} = q_0) \to 1.$$

Proof. For underfitting, Assumption 5 gives

$$\inf_{q<q_0} \{RSS_K(q) - RSS_K(q_0)\} \geq cT\Delta_T^2 + o_p(T\Delta_T^2).$$

The penalty difference for a fixed lower order is $o(T\Delta_T^2)$, so the deterministic loss from omitting a true break dominates any penalty saving. Thus underfitted orders are rejected with probability tending to one.

For overfitting,

$$\sup_{q>q_0} \{RSS_K(q_0) - RSS_K(q)\} = O_p(K \log T).$$

The penalty increment satisfies $K \log T = o\{\text{pen}_T(q, K)\}$, so any spurious RSS gain is dominated by the penalty increase with probability tending to one. Thus overfitted orders are rejected with probability tending to one. Combining the underfit and overfit cases gives $\Pr(\widehat{q} = q_0) \to 1$. $\square$

J.5 Gram stability

Lemma 11 (Gram stability gives well-behaved projections)**.** *Under Assumption 2, uniformly over $\Lambda \in \mathcal{L}_T$,*

$$\lambda_{\min}\{T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda}\} \geq c/2$$

with probability tending to one. Hence the coordinate formula for $\Pi_{K,\Lambda,T}$ is stable on $\mathcal{L}_T$, and the residual-maker $\boldsymbol{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$ remains an orthogonal contraction in $L^2(P_T)$:

$$\|\boldsymbol{M}_{K,\Lambda}\boldsymbol{z}\|_T \leq \|\boldsymbol{z}\|_T \qquad \text{for every } \boldsymbol{z} \in L^2(P_T).$$

Proof. By Weyl's eigenvalue inequality,

$$\lambda_{\min}\{T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda}\} \geq \lambda_{\min}\{G_Q(\Lambda)\} - \left\|T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda} - G_Q(\Lambda)\right\|_{op}.$$

Taking the infimum over $\Lambda \in \mathcal{L}_T$, Assumption 2 puts the first term above c and the second term below $c/2$ with probability tending to one. This proves the eigenvalue bound.

On that event, the least-squares projection has the usual stable coordinate form

$$\Pi_{K,\Lambda,T} = \boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda})^{-1}\boldsymbol{Q}'_{K,\Lambda}.$$

Independently of coordinates, $\Pi_{K,\Lambda,T}$ is the orthogonal projection onto $\mathcal{N}_{K,\Lambda,T}$. Therefore $\boldsymbol{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$ is also an orthogonal projection. Orthogonal projections are contractions in Hilbert norm. $\square$

J.6 Empirical-to-population projection bridge

Lemma 12 (Inner-product bridge from $L^2(P_T)$ to $L^2(P)$)**.** *Under Assumption 2, uniformly over the sieve and break classes used in the projected score,*

$$\langle \boldsymbol{f}_T, \boldsymbol{g}_T \rangle_T = \langle f, g \rangle_P + o_p(1).$$

In particular,

$$\sup_{\Lambda \in \mathcal{L}_T} \left\| T^{-1} \boldsymbol{Q}'_{K,\Lambda} \boldsymbol{Q}_{K,\Lambda} - G_Q(\Lambda) \right\|_{op} = o_p(1),$$

and the corresponding projected (x, w, u) cross moments converge to their population limits.

Proof. The first display is the uniform empirical law in Assumption 2. Taking f and g to be two coordinates of the block sieve row gives the displayed Gram convergence. Taking one or both functions to be the projected x, w, or u directions gives the corresponding cross-moment convergence. This is exactly the bridge needed by the workbook: finite-sample projection geometry is written in $L^2(P_T)$, while identification is written in $L^2(P)$, and the common object is convergence of the relevant inner products. □

J.7 Sieve remainder is small

Lemma 13 (True-partition sieve remainder is negligible)**.** *Under Assumption 3,*

$$\langle \boldsymbol{M}_0 \boldsymbol{\mu}_0, \boldsymbol{v}_0 \rangle_T = o_p(1), \qquad \sqrt{T} \langle \boldsymbol{M}_0 \boldsymbol{\mu}_0, \boldsymbol{v}_0 \rangle_T = o_p(1),$$

and

$$V_T^0 = P_T\{(\boldsymbol{M}_0 \boldsymbol{u})_t v_{0,t}\}^2 + o_p(1).$$

Proof. By definition $\boldsymbol{r}_{\mu,T} = \boldsymbol{M}_0 \boldsymbol{\mu}_0$. The first two displays are exactly the consistency-scale and score-scale components of Assumption 3.

At β_0,

$$\boldsymbol{M}_0 \boldsymbol{y}^\circ_{\beta_0} = \boldsymbol{M}_0(\boldsymbol{\mu}_0 + \boldsymbol{u}) = \boldsymbol{r}_{\mu,T} + \boldsymbol{M}_0 \boldsymbol{u}.$$

Thus

$$g_t(\beta_0, \Lambda_0) = \{(\boldsymbol{M}_0 \boldsymbol{u})_t + r_{\mu,T,t}\} v_{0,t}.$$

Let $\tilde{g}_t^0 = (\boldsymbol{M}_0 \boldsymbol{u})_t v_{0,t}$ and $d_t = r_{\mu,T,t} v_{0,t}$. Then

$$V_T^0 - P_T(\tilde{g}_t^0)^2 = 2 P_T \tilde{g}_t^0 d_t + P_T d_t^2.$$

The final term is $o_p(1)$ by Assumption 3. The cross term is bounded by Cauchy-Schwarz:

$$|P_T \tilde{g}_t^0 d_t| \le \{P_T(\tilde{g}_t^0)^2\}^{1/2} \{P_T d_t^2\}^{1/2} = o_p(1),$$

using the bounded second moment in Assumption 7. Hence $V_T^0 = P_T(\tilde{g}_t^0)^2 + o_p(1)$. □

J.8 Break replacement

Lemma 14 (Estimated partition may replace the true partition)**.** *Under Assumption 4,*

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\widehat{\Lambda}) - \Psi_T(\beta,\Lambda_0)| = o_p(1),$$

$$\sqrt{T}\,|\Psi_T(\beta_0,\widehat{\Lambda}) - \Psi_T(\beta_0,\Lambda_0)| = o_p(1), \qquad \widehat{V}_T - V_T^0 = o_p(1).$$

Proof. By Assumption 4, $\Pr\{\widehat{\Lambda} \in \mathcal{N}_T(r_T)\} \to 1$. On this event, $\widehat{\Lambda}$ is one of the partitions over which the three local suprema in the same assumption are taken. Therefore

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\widehat{\Lambda}) - \Psi_T(\beta,\Lambda_0)| \le \sup_{\Lambda\in\mathcal{N}_T(r_T)} \sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\Lambda) - \Psi_T(\beta,\Lambda_0)| = o_p(1),$$

$$\sqrt{T}|\Psi_T(\beta_0,\widehat{\Lambda}) - \Psi_T(\beta_0,\Lambda_0)| \le \sup_{\Lambda\in\mathcal{N}_T(r_T)} \sqrt{T}|\Psi_T(\beta_0,\Lambda) - \Psi_T(\beta_0,\Lambda_0)| = o_p(1),$$

and

$$|\widehat{V}_T - V_T^0| \le \sup_{\Lambda\in\mathcal{N}_T(r_T)} |V_T(\Lambda) - V_T^0| = o_p(1).$$

The conclusion follows after adding the negligible probability of the complement of the localization event. □

J.9 Uniform moment bridge

Lemma 15 (Uniform projected moment bridge)**.** *Under Assumptions 3, 4, and 6,*

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\widehat{\Lambda}) - \Psi(\beta,\Lambda_0)| = o_p(1). \tag{J.1}$$

Proof. First prove the bridge at the true partition. Since $\boldsymbol{y}_\beta^\circ = \boldsymbol{\mu}_0 + \boldsymbol{u} - (\beta-\beta_0)\boldsymbol{x}$,

$$\begin{aligned}\Psi_T(\beta,\Lambda_0) &= \langle \boldsymbol{M}_0\boldsymbol{y}_\beta^\circ, \boldsymbol{v}_0\rangle_T \\ &= \langle \boldsymbol{M}_0\boldsymbol{\mu}_0, \boldsymbol{v}_0\rangle_T + \langle \boldsymbol{M}_0\boldsymbol{u}, \boldsymbol{v}_0\rangle_T - (\beta-\beta_0)\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_0\rangle_T.\end{aligned}$$

The population moment from Part I is

$$\Psi(\beta,\Lambda_0) = \langle M_0^P U, v_0^P\rangle_P - (\beta-\beta_0)\langle M_0^P X, v_0^P\rangle_P.$$

Subtracting gives

$$\begin{aligned}\Psi_T(\beta,\Lambda_0) - \Psi(\beta,\Lambda_0) = {}& \langle \boldsymbol{M}_0\boldsymbol{\mu}_0, \boldsymbol{v}_0\rangle_T \\ &+ \left\{\langle \boldsymbol{M}_0\boldsymbol{u}, \boldsymbol{v}_0\rangle_T - \langle M_0^P U, v_0^P\rangle_P\right\} \\ &- (\beta-\beta_0)\left\{\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_0\rangle_T - \langle M_0^P X, v_0^P\rangle_P\right\}.\end{aligned}$$

The first term is $o_p(1)$ by Lemma 13. The next two bracketed terms are $o_p(1)$ by Assumption 6. Because $\mathcal{B}$ is fixed and bounded, the convergence is uniform in $\beta \in \mathcal{B}$:

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\Lambda_0) - \Psi(\beta,\Lambda_0)| = o_p(1).$$

Finally, Lemma 14 gives

$$\sup_{\beta\in\mathcal{B}} |\Psi_T(\beta,\widehat{\Lambda}) - \Psi_T(\beta,\Lambda_0)| = o_p(1).$$

The triangle inequality proves the stated bridge. □

J.10 Score CLT

Lemma 16 (Projected score CLT)**.** *Under Assumptions 3, 4, and 6,*

$$\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda}) \Rightarrow N(0, V).$$

Proof. At the true partition,

$$\begin{aligned}\sqrt{T}\Psi_T(\beta_0, \Lambda_0) &= \sqrt{T}\langle \boldsymbol{M}_0 \boldsymbol{y}^{\circ}_{\beta_0}, \boldsymbol{v}_0\rangle_T \\ &= T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0 + \sqrt{T}\langle \boldsymbol{M}_0\boldsymbol{\mu}_0, \boldsymbol{v}_0\rangle_T.\end{aligned}$$

The first equality is the definition of Ψ_T. The second uses $\boldsymbol{y}^{\circ}_{\beta_0} = \boldsymbol{\mu}_0 + \boldsymbol{u}$ and the symmetry/idempotence of $\boldsymbol{M}_0$, so $\langle \boldsymbol{M}_0\boldsymbol{u}, \boldsymbol{v}_0\rangle_T = T^{-1}\boldsymbol{u}'\boldsymbol{v}_0$. The approximation-bias term is $o_p(1)$ by Assumption 3. The leading term $T^{-1/2}\boldsymbol{u}'\boldsymbol{v}_0$ converges to $N(0, V)$ by Assumption 6. Therefore

$$\sqrt{T}\Psi_T(\beta_0, \Lambda_0) \Rightarrow N(0, V).$$

Lemma 14 gives

$$\sqrt{T}\{\Psi_T(\beta_0, \widehat{\Lambda}) - \Psi_T(\beta_0, \Lambda_0)\} = o_p(1).$$

Slutsky's theorem gives the result for $\widehat{\Lambda}$. □

J.11 Variance consistency

Lemma 17 (Projected variance consistency)**.** *Under Assumptions 3, 4, and 7,*

$$\widehat{V}_T \to_p V > 0.$$

Proof. Lemma 13 gives

$$V_T^0 = P_T(\tilde{g}_t^0)^2 + o_p(1), \qquad \tilde{g}_t^0 = (\boldsymbol{M}_0\boldsymbol{u})_t v_{0,t}.$$

Assumption 7 gives $P_T(\tilde{g}_t^0)^2 \to_p V$. Therefore $V_T^0 \to_p V$. Lemma 14 gives $\widehat{V}_T - V_T^0 = o_p(1)$, so $\widehat{V}_T \to_p V$. The lower bound $V > 0$ is part of Assumption 6; it rules out a degenerate residual score direction. □

J.12 Max-score and convex hull

Lemma 18 (EL local feasibility and max-score)**.** *Under Assumptions 7 and 8,*

$$\max_{t\le T} |g_t(\beta_0, \Lambda_0)|/\sqrt{T} = o_p(1), \qquad \max_{t\le T} |g_t(\beta_0, \widehat{\Lambda})|/\sqrt{T} = o_p(1),$$

and the origin is in the interior of the scalar convex hull for both $\{g_t(\beta_0, \Lambda_0) : 1 \le t \le T\}$ *and* $\{g_t(\beta_0, \widehat{\Lambda}) : 1 \le t \le T\}$ *with probability tending to one.*

Proof. The two max-score bounds are displayed in Assumption 7. For scalar moments, the convex hull of the observed values is the interval from the sample minimum to the sample maximum. The origin is in its interior exactly when the minimum is strictly negative and the maximum is strictly positive. Assumption 8 states that this sign condition holds with probability tending to one for both the known-partition and estimated-partition moment arrays. □

J.13 EL KKT local expansion

Lemma 19 (Local EL KKT expansion)**.** *Under Assumptions 3, 4, 6, 7, and 8, with $g_t = g_t(\beta_0, \widehat{\Lambda})$,*

$$\lambda_T = \frac{P_T g_t}{P_T g_t^2} + o_p(T^{-1/2}), \qquad \max_{t \le T} |\lambda_T g_t| = o_p(1),$$

and

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{\widehat{V}_T} + o_p(1).$$

Proof. By Lemma 18, the scalar convex-hull condition holds with probability tending to one, so the EL dual root λ_T exists in the interior region where $1 + \lambda_T g_t > 0$. By Lemma 16,

$$P_T g_t = \Psi_T(\beta_0, \widehat{\Lambda}) = O_p(T^{-1/2}).$$

By Lemma 17,

$$P_T g_t^2 = \widehat{V}_T \to_p V > 0.$$

Expanding the KKT equation

$$0 = P_T\left[\frac{g_t}{1 + \lambda_T g_t}\right]$$

around $\lambda_T = 0$ gives

$$0 = P_T g_t - \lambda_T P_T g_t^2 + O_p\{\lambda_T^2 P_T |g_t|^3\}.$$

The third-moment bound in Assumption 7 implies the remainder is $O_p(\lambda_T^2)$. Since $P_T g_t = O_p(T^{-1/2})$ and $P_T g_t^2 \to_p V > 0$, the root satisfies $\lambda_T = O_p(T^{-1/2})$ and hence

$$\lambda_T = \frac{P_T g_t}{P_T g_t^2} + o_p(T^{-1/2}).$$

Combining $\lambda_T = O_p(T^{-1/2})$ with the max-score bound gives

$$\max_{t \le T} |\lambda_T g_t| \le |\lambda_T| \max_{t \le T} |g_t| = o_p(1).$$

Therefore the log EL ratio can be Taylor-expanded:

$$\begin{aligned}
\ell_T(\beta_0, \widehat{\Lambda}) &= 2\sum_{t=1}^{T} \log(1 + \lambda_T g_t) \\
&= 2T\lambda_T P_T g_t - T\lambda_T^2 P_T g_t^2 + O_p\{T|\lambda_T|^3 P_T |g_t|^3\}.
\end{aligned}$$

The final term is $o_p(1)$, because $\lambda_T = O_p(T^{-1/2})$ and $P_T|g_t|^3 = O_p(1)$. Substituting $\lambda_T = P_T g_t / P_T g_t^2 + o_p(T^{-1/2})$ yields

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T(P_T g_t)^2}{P_T g_t^2} + o_p(1) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{\widehat{V}_T} + o_p(1).$$

□

J.14 Known-partition Wilks

Proposition 7 (Known-partition profile EL Wilks)**.** *Under the true partition Λ_0, the profile empirical likelihood ratio satisfies*

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \chi_1^2.$$

Proof. The true-partition versions of Lemmas 16 and 17 give

$$\sqrt{T}\Psi_T(\beta_0, \Lambda_0) \Rightarrow N(0, V), \qquad V_T^0 \to_p V > 0.$$

The max-score and convex-hull parts of Assumptions 7 and 8 apply to $g_t(\beta_0, \Lambda_0)$, so the KKT expansion from Part I gives

$$\ell_T(\beta_0, \Lambda_0) = \frac{T\{\Psi_T(\beta_0, \Lambda_0)\}^2}{V_T^0} + o_p(1).$$

Therefore

$$\ell_T(\beta_0, \Lambda_0) = \left\{ \frac{\sqrt{T}\Psi_T(\beta_0, \Lambda_0)}{\sqrt{V_T^0}} \right\}^2 + o_p(1) \Rightarrow \chi_1^2.$$

□

J.15 Part II consequences

The uniform bridge gives the consistency input. If $\widehat{\beta}_T$ is an approximate sample root,

$$\Psi_T(\widehat{\beta}_T, \widehat{\Lambda}) = o_p(1),$$

then Lemma 15 gives

$$\Psi(\widehat{\beta}_T, \Lambda_0) = o_p(1).$$

Part I gives the unique population zero, so $\widehat{\beta}_T \to_p \beta_0$.

For Wilks, Lemmas 16, 17, and 19 give

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{\widehat{V}_T} + o_p(1), \qquad \frac{\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda})}{\sqrt{\widehat{V}_T}} \Rightarrow N(0, 1).$$

Part III only merges these two displayed statements with the geometric KKT reduction already proved in Part I.

Part IV
Merging Part

K Merging Part: Moment Bridge, EL KKT, and χ_1^2

The merge step should not say that the unscaled moment bridge is itself equivalent to Wilks. The correct statement is more precise:

The same projected moment Ψ_T is the interface for both targets. Unscaled convergence of Ψ_T to Ψ merges with population uniqueness to prove consistency. Scaled convergence of Ψ_T, together with the EL KKT quadratic reduction, merges to prove χ_1^2.

K.1 Merge A: consistency

Part I gives the population geometry

$$\Psi(\beta, \Lambda_0) = \langle M_0^P Y_\beta^\circ, v_0^P \rangle_P = -(\beta - \beta_0)\Gamma$$

under exogeneity and relevance. Hence

$$\Psi(\beta, \Lambda_0) = 0 \quad \Longleftrightarrow \quad \beta = \beta_0.$$

Part II gives the stochastic bridge

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \widehat{\Lambda}) - \Psi(\beta, \Lambda_0)| = o_p(1).$$

Therefore any approximate sample root satisfies

$$\Psi_T(\widehat{\beta}_T, \widehat{\Lambda}) = o_p(1) \quad \Longrightarrow \quad \widehat{\beta}_T \to_p \beta_0.$$

This is the estimation target.

K.2 Merge B: Wilks inference

For inference, Part I gives the EL KKT representation

$$\ell_T(\beta, \widehat{\Lambda}) = 2 \sum_{t=1}^{T} \log\{1 + \lambda_T g_t(\beta, \widehat{\Lambda})\},$$

where λ_T solves

$$P_T \left[\frac{g_t(\beta, \widehat{\Lambda})}{1 + \lambda_T g_t(\beta, \widehat{\Lambda})} \right] = 0.$$

Part II verifies the local size and convex-hull conditions, so the KKT equation can be expanded at $\beta = \beta_0$:

$$\ell_T(\beta_0, \widehat{\Lambda}) = \frac{T\{\Psi_T(\beta_0, \widehat{\Lambda})\}^2}{V_T(\widehat{\Lambda})} + o_p(1).$$

Part II also gives

$$\frac{\sqrt{T}\Psi_T(\beta_0, \widehat{\Lambda})}{\sqrt{V_T(\widehat{\Lambda})}} \Rightarrow N(0, 1).$$

Combining the last two displays gives

$$\ell_T(\beta_0, \widehat{\Lambda}) \Rightarrow \chi_1^2. \tag{K.1}$$

Thus the "equivalence" is conditional on the geometric KKT reduction. Once EL has been reduced to the square of the standardized projected moment, Wilks is equivalent to the standardized score CLT. It is not equivalent to the unscaled LLN bridge used for consistency.

Proof-body split. Geometry owns the model, nuisance tangent space, projection KKT, residual score, population zero, and EL primal/dual KKT. Small stochastic bounds only verify convergence and local size. The merging part then turns the same projected moment into either consistency or Wilks.

L Optional Semiparametric Efficiency Theorem With Efficient Weight

This optional block comes after the Wilks argument and is not part of the minimum profile-sieve Wilks proof. The generic projected score with a residualized instrument w gives valid orthogonal EL inference under the stochastic conditions above. A Fisher-style semiparametric efficiency claim needs an extra local likelihood model and the efficient residualized instrument.

The logic is: choose $w = w^*$, prove efficient-score equivalence, then derive the asymptotic linear expansion. The homoskedastic case reduces to $w = X$; the heteroskedastic oracle case uses $w^* = X/\sigma^2(A)$.

Let

$$A = (J_0, W),$$

where J_0 is the true regime indicator. If the paper keeps the notation $A = (S, W)$, then every conditional projection $E(\cdot \mid A)$ below should be read as projection onto the true break-aware nuisance sigma-field generated by the true regime and W. This avoids treating the nuisance space as an arbitrary continuous-time S-indexed space. The population model is

$$Y = \beta_0 X + \mu_0(A) + U, \qquad \mu_0(A) \in \mathcal{N}_0^P = \mathcal{N}_{\Lambda_0}^P.$$

For the efficiency calculation, assume the conditional Gaussian location model

$$U \mid X, A \sim N(0, \sigma^2(A)), \qquad 0 < c \le \sigma^2(A) \le C < \infty.$$

The homoskedastic case is the special case $\sigma^2(A) = \sigma^2$. The population nuisance tangent directions generated by paths $\mu_t(A) = \mu_0(A) + th(A)$, $h \in \mathcal{N}_0^P$, are

$$\mathcal{T}_\mu = \overline{\left\{ \frac{Uh(A)}{\sigma^2(A)} : h \in \mathcal{N}_0^P \right\}}^{L^2(P)}.$$

The raw score for β is

$$S_\beta^0 = \frac{UX}{\sigma^2(A)}.$$

Proposition 8 (Efficient score)**.** *Let*

$$m_X(A) = E(X \mid A), \qquad \widetilde{X} = X - m_X(A).$$

The efficient score for β is

$$S_{\text{eff}} = \frac{U\widetilde{X}}{\sigma^2(A)},$$

and the efficient information is

$$I_{\text{eff}} = E(S_{\text{eff}}^2) = E\left\{\frac{\widetilde{X}^2}{\sigma^2(A)}\right\}.$$

Proof. The projection of S_β^0 onto $\mathcal{T}_\mu$ has the form $Uh^*(A)/\sigma^2(A)$. The function h^* minimizes

$$E\left[\left\{\frac{U}{\sigma^2(A)}\right\}^2 \{X - h(A)\}^2\right].$$

Since $E(U^2 \mid X, A) = \sigma^2(A)$, this criterion is

$$E\left[\frac{\{X - h(A)\}^2}{\sigma^2(A)}\right].$$

Because $\sigma^2(A)$ is A-measurable, conditional minimization gives $h^*(A) = E(X \mid A) = m_X(A)$. Therefore

$$\Pi(S_\beta^0 \mid \mathcal{T}_\mu) = \frac{Um_X(A)}{\sigma^2(A)},$$

and

$$S_{\text{eff}} = S_\beta^0 - \Pi(S_\beta^0 \mid \mathcal{T}_\mu) = \frac{U\{X - m_X(A)\}}{\sigma^2(A)}.$$

Finally,

$$E(S_{\text{eff}}^2) = E\left\{\frac{U^2\widetilde{X}^2}{\sigma^4(A)}\right\} = E\left\{\frac{\widetilde{X}^2}{\sigma^2(A)}\right\},$$

again using $E(U^2 \mid X, A) = \sigma^2(A)$. $\square$

The efficient population weight is

$$w^*(O) = \frac{X}{\sigma^2(A)}.$$

Since $\sigma^2(A)$ is A-measurable,

$$E(w^* \mid A) = \frac{E(X \mid A)}{\sigma^2(A)}.$$

Thus the population residualized efficient instrument is

$$M_0^P w^* = w^* - E(w^* \mid A) = \frac{\widetilde{X}}{\sigma^2(A)}. \tag{L.1}$$

At β_0, $Y_{\beta_0}^\circ = Y - \beta_0 X = \mu_0(A) + U$, and $M_0^P \mu_0 = 0$. Hence

$$M_0^P Y_{\beta_0}^\circ \, M_0^P w^* = U\frac{\widetilde{X}}{\sigma^2(A)} = S_{\text{eff}}.$$

This is the population reason the profile score becomes efficient when $w = w^*$. If $\sigma^2(A)$ is constant, multiplying the moment by the constant $1/\sigma^2$ does not change the root, so $w = X$ is the efficient choice in the homoskedastic case.

For the sample statement, define

$$w_t^* = \frac{x_t}{\sigma^2(A_t)}, \qquad \boldsymbol{w}^* = (w_1^*, \dots, w_T^*)', \qquad \boldsymbol{v}_{K,T}^* = \boldsymbol{M}_0 \boldsymbol{w}^*,$$

where $A_t = (J_{0,t}, W_{t-1})$. Here $J_{0,t} = j$ iff $k_{j,0} < t \le k_{j+1,0}$, and $\boldsymbol{M}_0 = \boldsymbol{M}_{K,\Lambda_0}$. The efficient sieve approximation target is

$$\left\| \boldsymbol{v}_{K,T}^* - \left(\frac{\widetilde{X}_1}{\sigma^2(A_1)}, \dots, \frac{\widetilde{X}_T}{\sigma^2(A_T)} \right)' \right\|_T = o_p(1). \tag{L.2}$$

This follows from the same finite-sieve approximation and Gram-stability logic used in Part II, provided the block sieve approximates $E(w^* \mid A) = E(X \mid A)/\sigma^2(A)$.

Lemma 20 (Efficient-score equivalence)**.** *Assume* (L.2) *and the true-partition efficient-weight bias condition*

$$T^{-1/2} \boldsymbol{r}_{\mu,T}' \boldsymbol{v}_{K,T}^* = o_p(1), \qquad \boldsymbol{r}_{\mu,T} = \boldsymbol{M}_0 \boldsymbol{\mu}_0.$$

Then

$$\sqrt{T}\Psi_T^*(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^{T} S_{\text{eff},t} + o_p(1),$$

where

$$\Psi_T^*(\beta, \Lambda) = \langle \boldsymbol{M}_{K,\Lambda}(\boldsymbol{y} - \beta \boldsymbol{x}), \boldsymbol{M}_{K,\Lambda} \boldsymbol{w}^* \rangle_T.$$

Proof. At β_0,

$$\boldsymbol{y}_{\beta_0}^{\circ} = \boldsymbol{\mu}_0 + \boldsymbol{u}.$$

Therefore

$$\begin{aligned} \sqrt{T}\Psi_T^*(\beta_0, \Lambda_0) &= T^{-1/2} (\boldsymbol{M}_0 \boldsymbol{y}_{\beta_0}^{\circ})' \boldsymbol{v}_{K,T}^* \\ &= T^{-1/2} \boldsymbol{u}' \boldsymbol{v}_{K,T}^* + T^{-1/2} \boldsymbol{r}_{\mu,T}' \boldsymbol{v}_{K,T}^*. \end{aligned}$$

The second term is $o_p(1)$ by assumption. The first term equals

$$\begin{aligned} T^{-1/2} \boldsymbol{u}' \boldsymbol{v}_{K,T}^* &= T^{-1/2} \sum_{t=1}^{T} U_t \frac{\widetilde{X}_t}{\sigma^2(A_t)} + o_p(1) \\ &= T^{-1/2} \sum_{t=1}^{T} S_{\text{eff},t} + o_p(1). \end{aligned}$$

by (L.2) and the same Cauchy-Schwarz/variance argument used for the score replacement in Part II. $\square$

For fixed Λ, the efficient-weight moment is linear in β, with

$$\frac{\partial}{\partial\beta}\Psi_T^*(\beta,\Lambda) = -\langle \boldsymbol{M}_{K,\Lambda}\boldsymbol{x}, \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}^*\rangle_T.$$

Thus the derivative denominator in the root expansion is the empirical version of the efficient information.

Theorem 5 (Optional semiparametric efficiency with efficient weight)**.** *Let $\widehat{\beta}_T$ be a profile EL root, or an approximate profile moment root, satisfying*

$$\Psi_T^*(\widehat{\beta}_T, \widehat{\Lambda}) = o_p(T^{-1/2}).$$

Assume the efficient-score equivalence in Lemma 20, estimated-partition score equivalence at the $T^{-1/2}$ scale, and derivative equivalence:

$$\sqrt{T}\{\Psi_T^*(\beta_0, \widehat{\Lambda}) - \Psi_T^*(\beta_0, \Lambda_0)\} = o_p(1),$$

$$\langle \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{x}, \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{w}^*\rangle_T - \langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_{K,T}^*\rangle_T = o_p(1).$$

Assume also

$$\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_{K,T}^*\rangle_T \to_p I_{\text{eff}} > 0.$$

Then

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) = I_{\text{eff}}^{-1} T^{-1/2} \sum_{t=1}^{T} S_{\text{eff},t} + o_p(1),$$

and consequently

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) \Rightarrow N(0, I_{\text{eff}}^{-1}).$$

Thus the efficient-weight profile-sieve EL root attains the semiparametric efficiency bound.

Proof. By a mean-value expansion of the projected moment around β_0,

$$\begin{aligned}
0 &= \Psi_T^*(\widehat{\beta}_T, \widehat{\Lambda}) + o_p(T^{-1/2}) \\
&= \Psi_T^*(\beta_0, \widehat{\Lambda}) - (\widehat{\beta}_T - \beta_0)\langle \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{x}, \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{w}^*\rangle_T + o_p(T^{-1/2}) + o_p(|\widehat{\beta}_T - \beta_0|).
\end{aligned}$$

The score-equivalence and derivative-equivalence assumptions replace the estimated partition by the true partition. Hence

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) = \left\{\langle \boldsymbol{M}_0\boldsymbol{x}, \boldsymbol{v}_{K,T}^*\rangle_T\right\}^{-1} \sqrt{T}\Psi_T^*(\beta_0, \Lambda_0) + o_p(1).$$

The denominator converges to

$$E\{(M_0^P X)(M_0^P w^*)\} = E\left\{\widetilde{X}\frac{\widetilde{X}}{\sigma^2(A)}\right\} = I_{\text{eff}}.$$

Lemma 20 gives

$$\sqrt{T}\Psi_T^*(\beta_0, \Lambda_0) = T^{-1/2}\sum_{t=1}^{T} S_{\text{eff},t} + o_p(1).$$

This proves the asymptotic linear representation. Since $E(S_{\text{eff}} \mid X, A) = 0$ and $E(S_{\text{eff}}^2) = I_{\text{eff}}$, the relevant martingale or iid CLT gives

$$T^{-1/2} \sum_{t=1}^{T} S_{\text{eff},t} \Rightarrow N(0, I_{\text{eff}}).$$

Therefore

$$\sqrt{T}(\widehat{\beta}_T - \beta_0) \Rightarrow N(0, I_{\text{eff}}^{-1}).$$

The standard Cauchy-Schwarz information bound gives $E(\phi^2) \geq I_{\text{eff}}^{-1}$ for any regular influence function ϕ satisfying $E\{\phi S_{\text{eff}}\} = 1$. The influence function above is $I_{\text{eff}}^{-1} S_{\text{eff}}$, whose variance is I_{eff}^{-1}. It therefore attains the bound. □

Remark 2 (Estimated variance plug-in)**.** The main theorem is cleanest with known $\sigma^2(A)$, or with the homoskedastic simplification $w = X$. If $\sigma^2(A)$ is estimated, define

$$\widehat{w}_t^* = \frac{x_t}{\widehat{\sigma}^2(A_t)}, \qquad \widehat{\boldsymbol{w}}^* = (\widehat{w}_1^*, \ldots, \widehat{w}_T^*)'.$$

Replacing $\boldsymbol{w}^*$ by $\widehat{\boldsymbol{w}}^*$ leaves the same asymptotic linear representation if

$$\|\boldsymbol{M}_{K,\widehat{\Lambda}}(\widehat{\boldsymbol{w}}^* - \boldsymbol{w}^*)\|_T = o_p(1),$$

$$\sqrt{T}\left|\langle \boldsymbol{M}_{K,\widehat{\Lambda}}(\boldsymbol{y} - \beta_0 \boldsymbol{x}), \boldsymbol{M}_{K,\widehat{\Lambda}}(\widehat{\boldsymbol{w}}^* - \boldsymbol{w}^*)\rangle_T\right| = o_p(1),$$

and

$$\left|\langle \boldsymbol{M}_{K,\widehat{\Lambda}}\boldsymbol{x}, \boldsymbol{M}_{K,\widehat{\Lambda}}(\widehat{\boldsymbol{w}}^* - \boldsymbol{w}^*)\rangle_T\right| = o_p(1).$$

This plug-in statement is an appendix-level add-on; the main optional theorem should use either $w = X$ under homoskedasticity or the oracle heteroskedastic weight $w^* = X/\sigma^2(A)$.

This theorem is conditional on the efficiency model and the efficient-weight choice. With only martingale exogeneity and no conditional variance or local likelihood structure, the workbook proves valid profile EL Wilks inference, but not a Fisher-style semiparametric efficiency bound.